Imperial College London

Lecture 12

Reliability and Variations in VLSI

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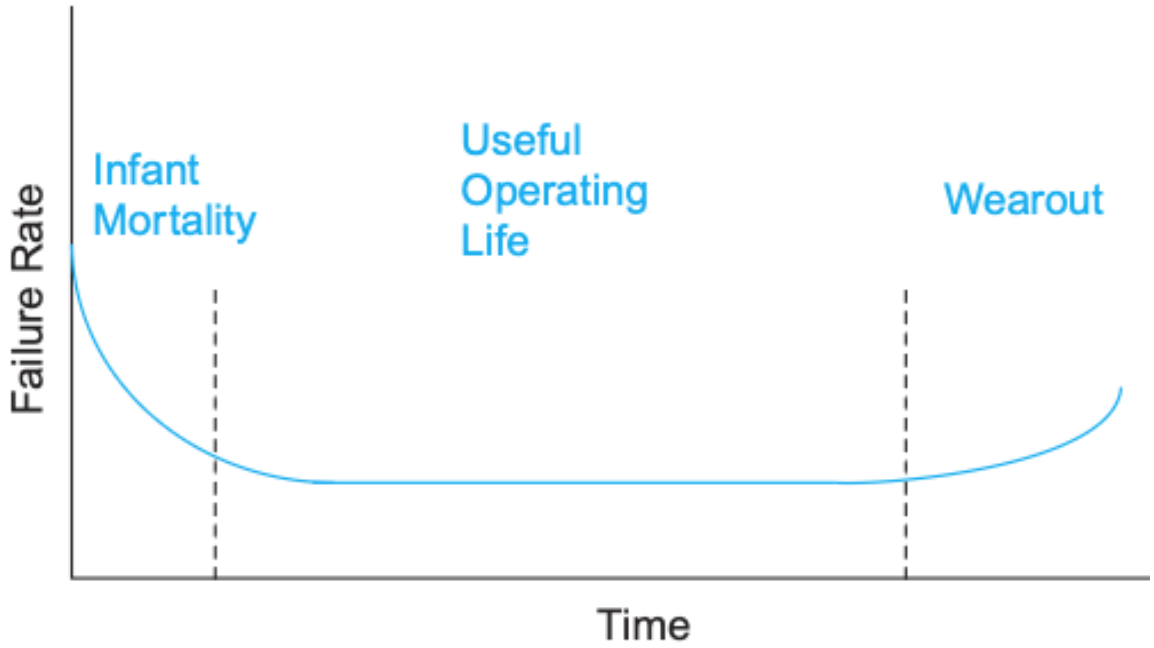
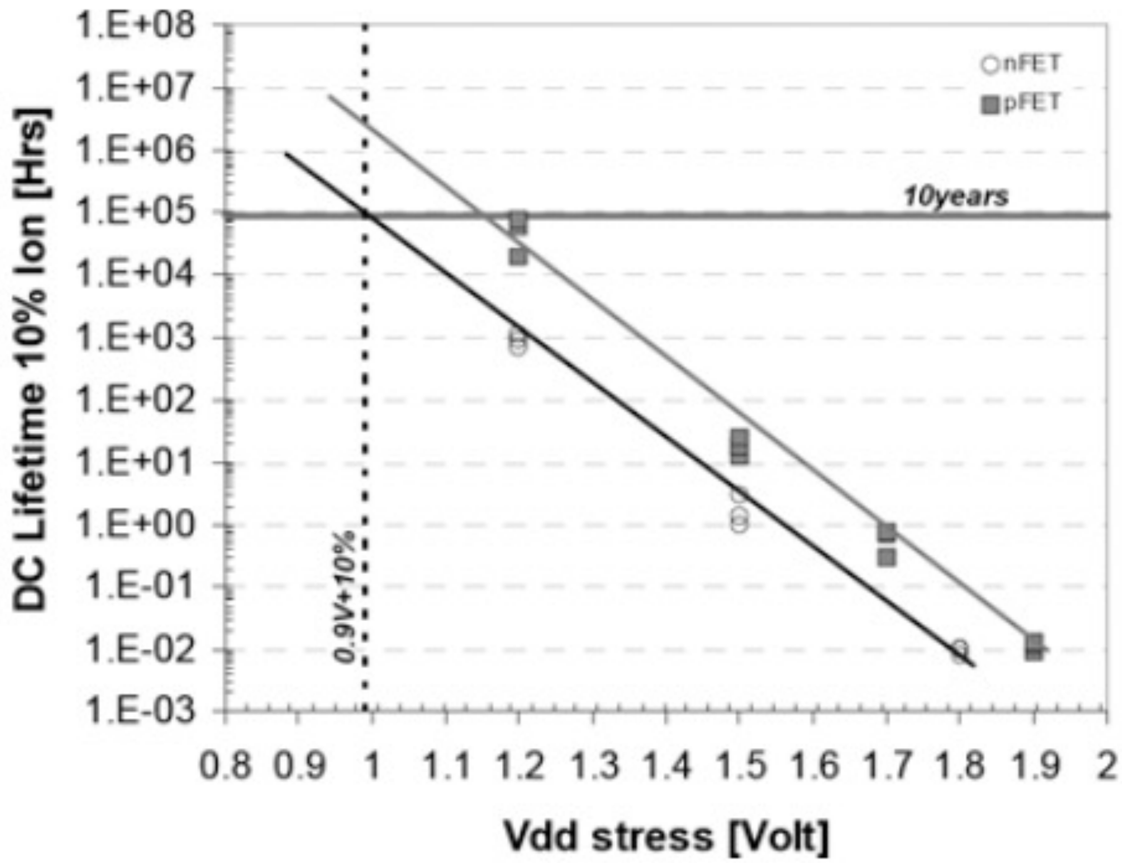
Reliability in VLSI

- Hard Errors
  - Oxide wearout
  - Interconnect wearout
  - Overvoltage failure
  - Latchup
- Soft Errors
- Characterizing reliability
  - Mean time between failures (MTBF)
    - # of devices x hours of operation / number of failures
  - Failures in time (FIT)
    - # of failures / thousand hours / million devices



Accelerated Lifetime Testing

- Expected reliability typically exceeds 10 years
- But products come to market in 1-2 years
- Accelerated lifetime testing required to predict adequate long-term reliability



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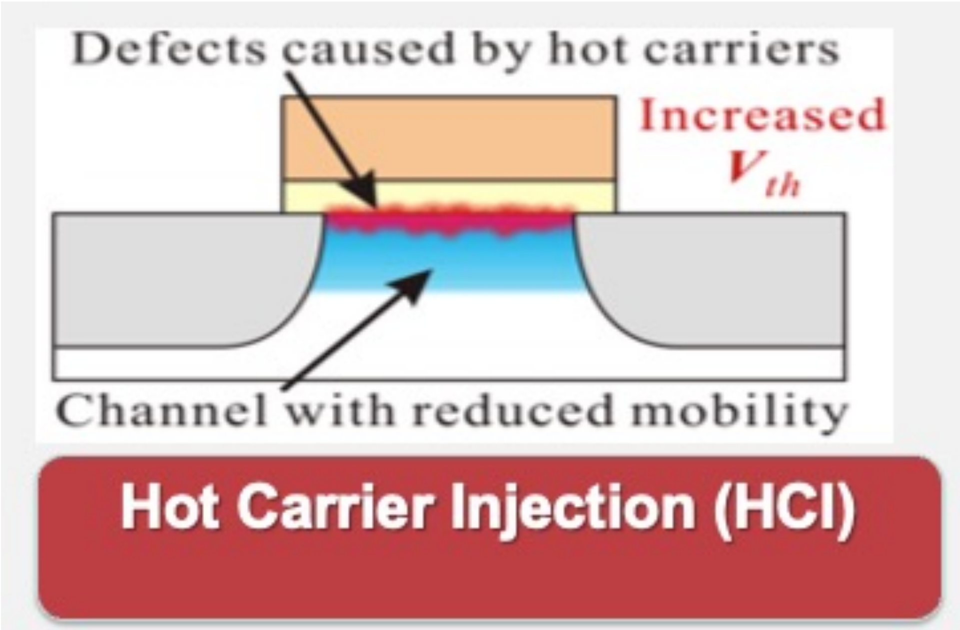
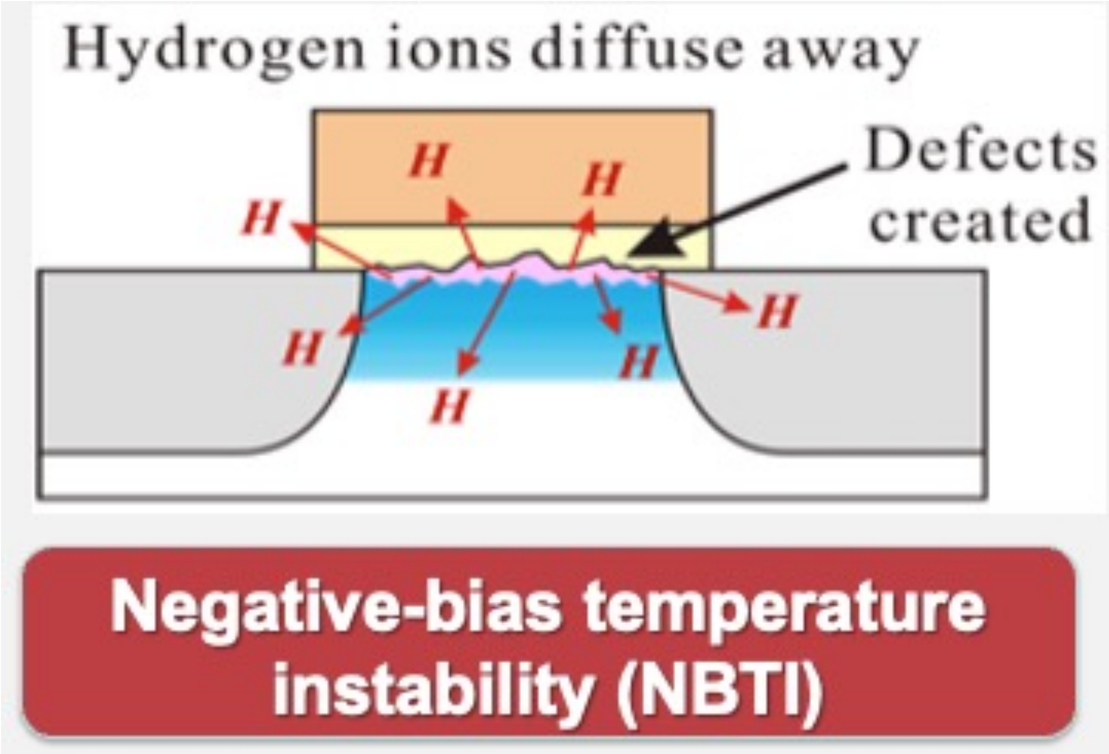
Hot Carriers

- Electric fields across channel impart high energies to some carriers
  - These "hot" carriers may be blasted into the gate oxide where they become trapped
  - Accumulation of charge in oxide causes shift in V_t over time
  - Eventually V_t shifts too far for devices to operate correctly
- Choose V_{DD} to achieve reasonable product lifetime
  - Worst problems for inverters and NORs with slow input risetime and long propagation delays



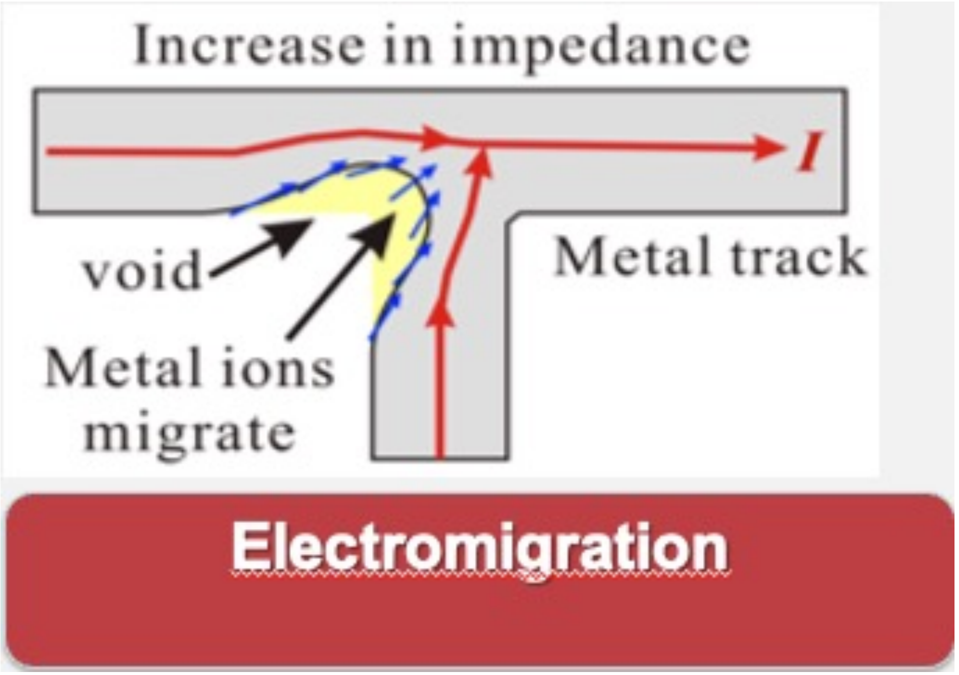
Negative-bias Temperature Instability (NBTI)

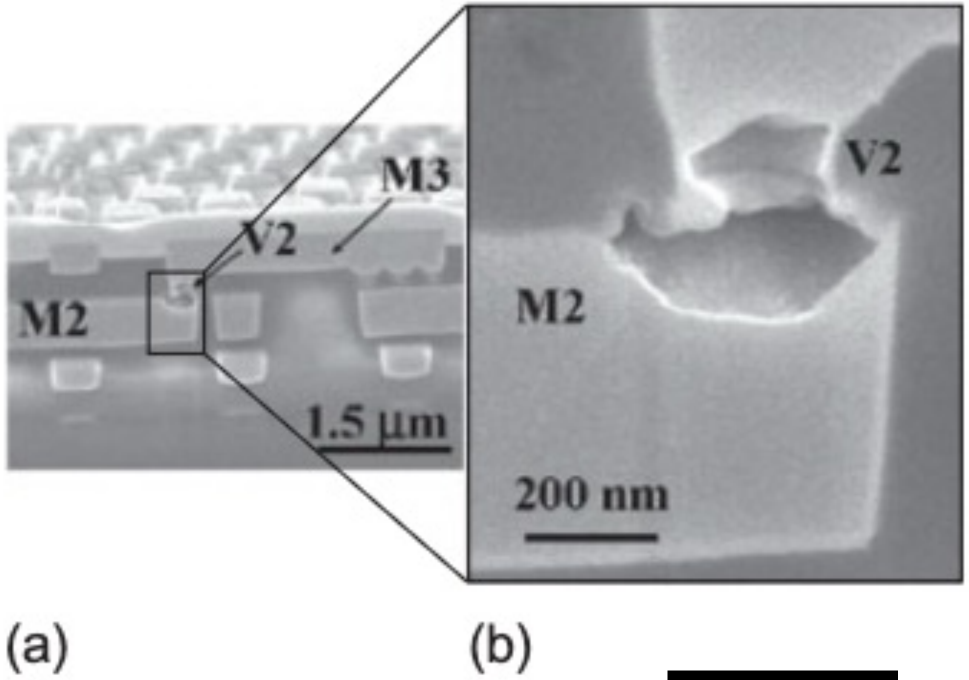
- Electric field applied across oxide forms dangling bonds called traps at Si-SiO_2 interface
- Accumulation of traps causes V_t shift
- Most pronounced for pMOS transistors with strong negative bias ($V_g = 0$, $V_s = V_{DD}$) at high temperature



Electromigration

- "Electron wind" causes movement of metal atoms along wires
- Excessive electromigration leads to open circuits
- Most significant for unidirectional (DC) current
  - Depends on current density J_{dc} (current / area)
  - Exponential dependence on temperature
- Typical limits: $J_{dc} < 1 – 2$ mA / μm^2





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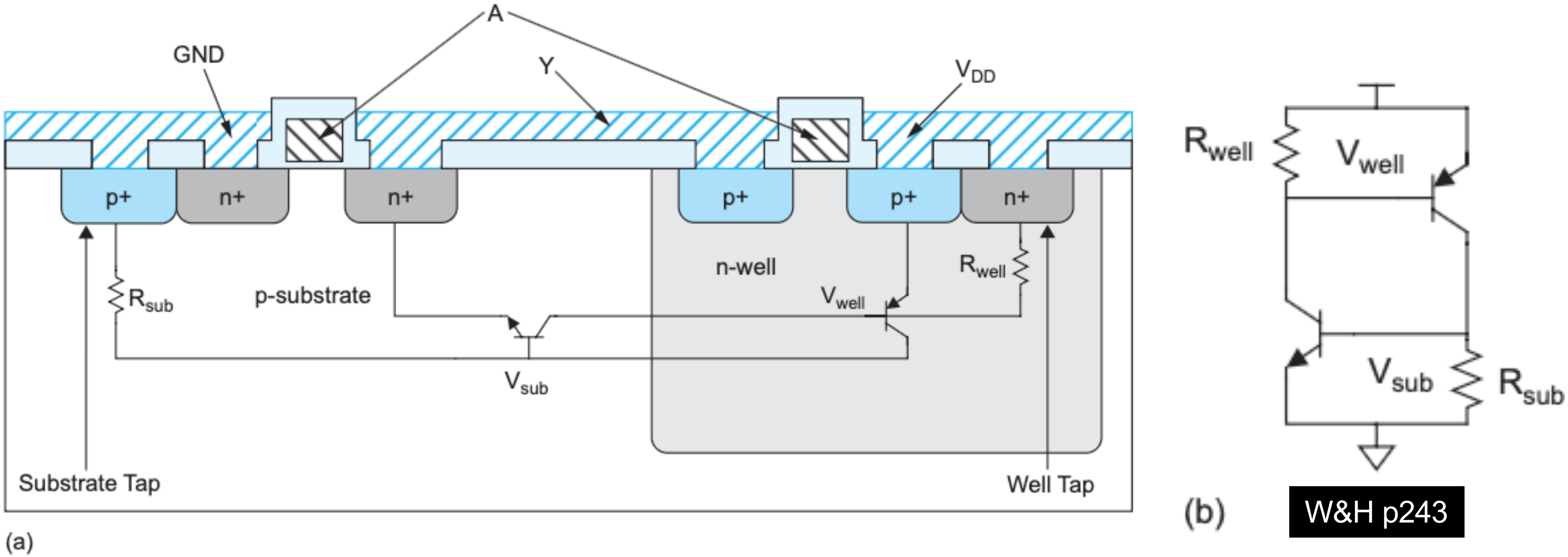
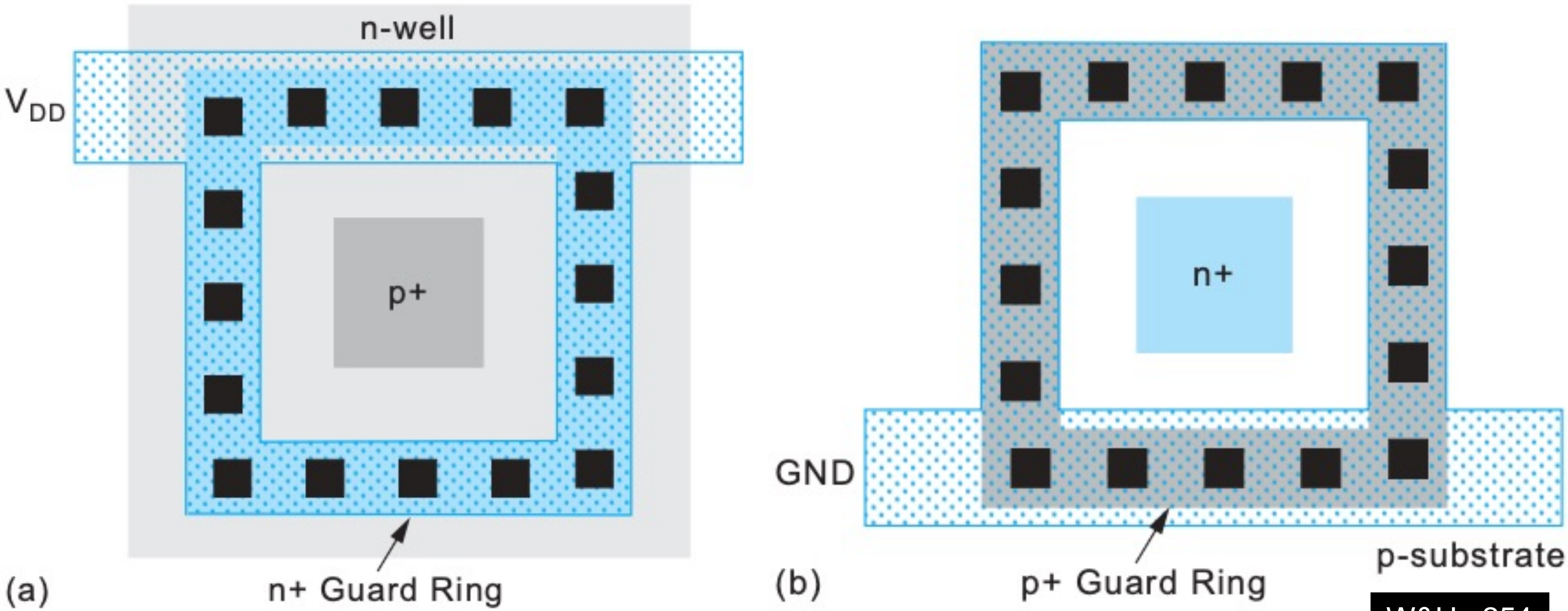
Latchup

- Latchup: positive feedback leading to V_{DD} – GND short
  - Major problem for 1970's CMOS processes before it was well understood
- Avoid by minimizing resistance of body to GND / V_{DD}
  - Use plenty of substrate and well taps



Latchup prevention with Guard Rings

- Latchup risk greatest when diffusion-to-substrate diodes could become forward-biased
- Surround sensitive region with guard ring to collect injected charge



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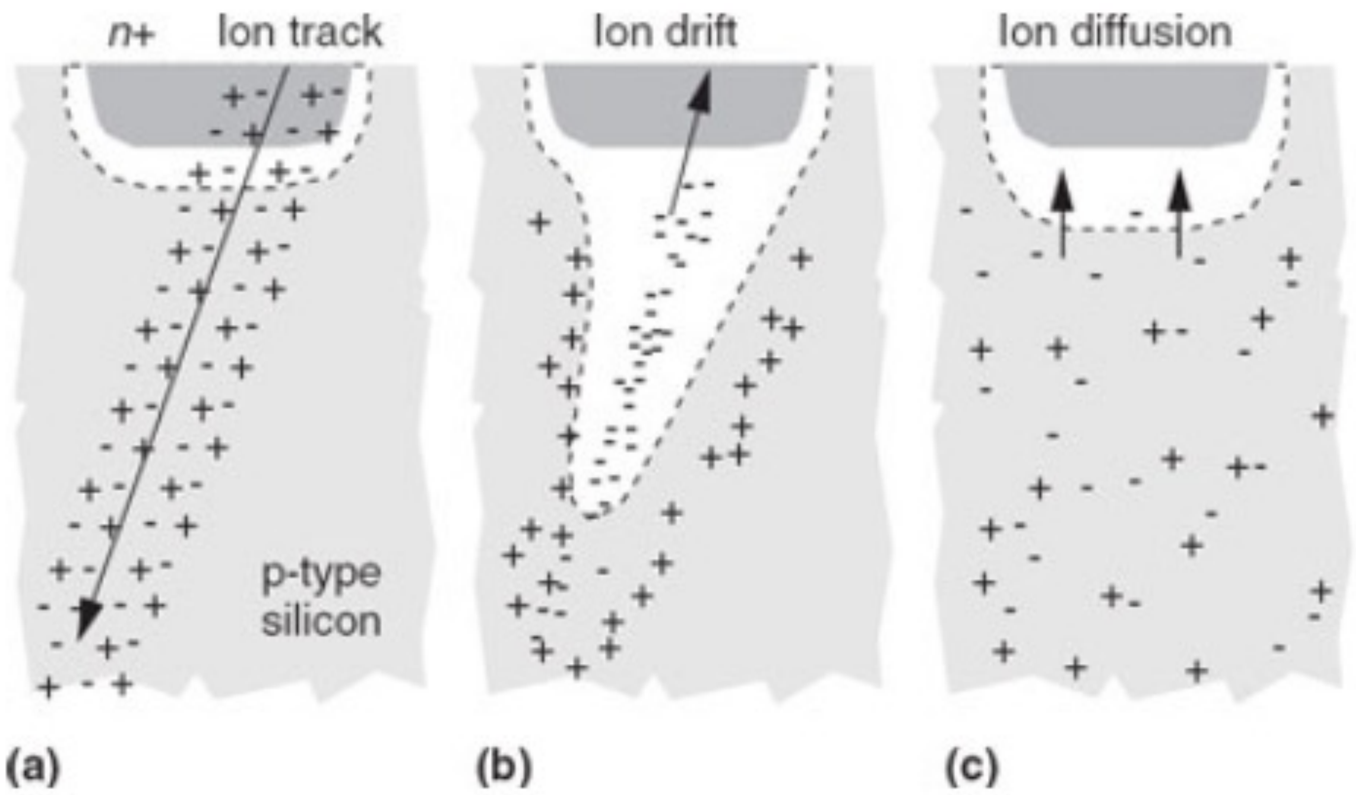
Soft Errors

- In 1970's, DRAMs were observed to randomly flip bits
  - Ultimately linked to alpha particles and cosmic ray neutrons
- Collisions with atoms create electron-hole pairs in substrate
  - These carriers are collected on p-n junctions, disturbing the voltage



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PVT Variability in VLSI chips

- There are three main causes :

1. **Process** variation affecting (P):
   - Threshold voltages
   - Channel lengths
   - Wire dimensions
2. Environment:
   - Operating **voltage** (V)
   - Operating **temperature** (T)
3. Aging or wearout
   - Degradation
   - Electron migration

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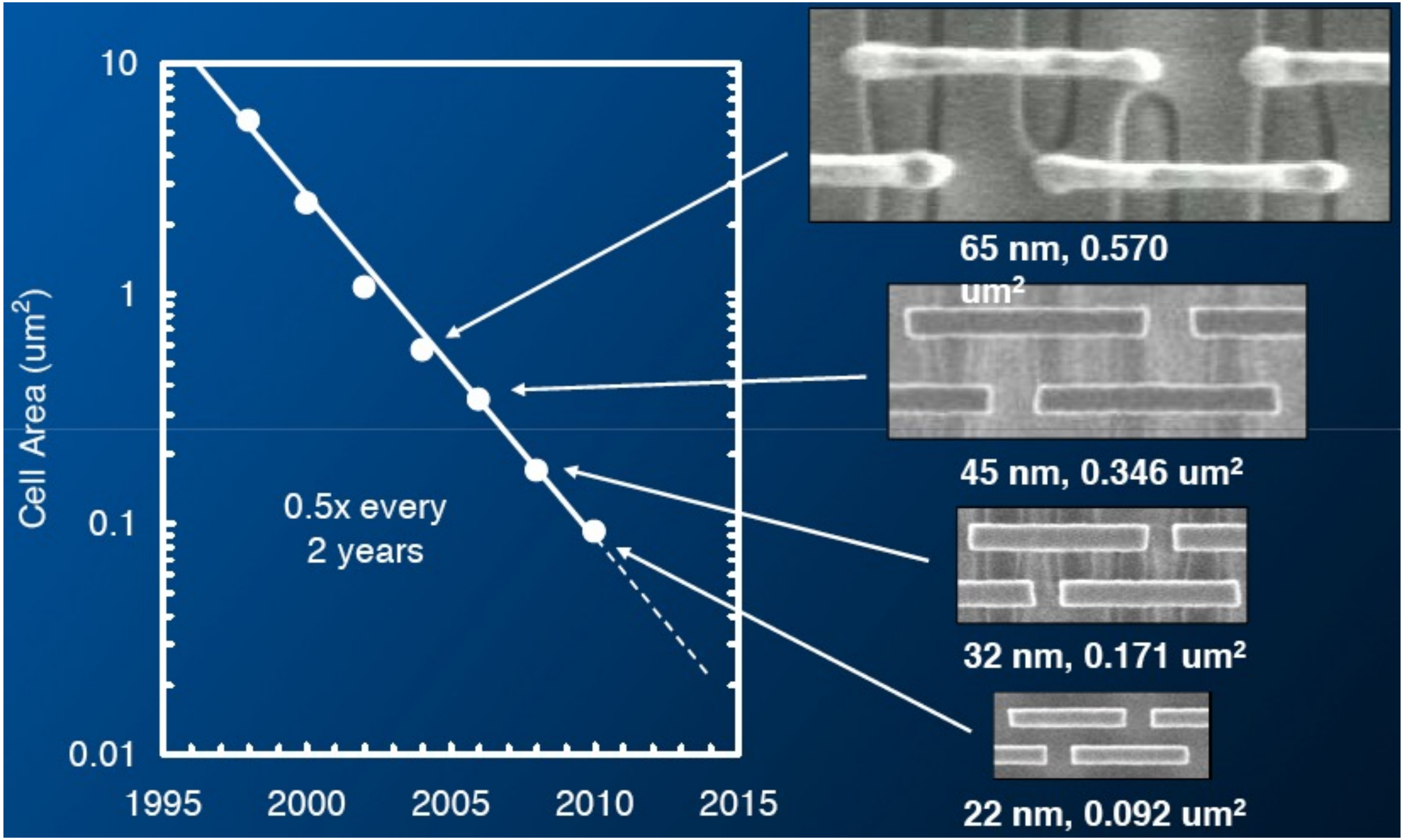
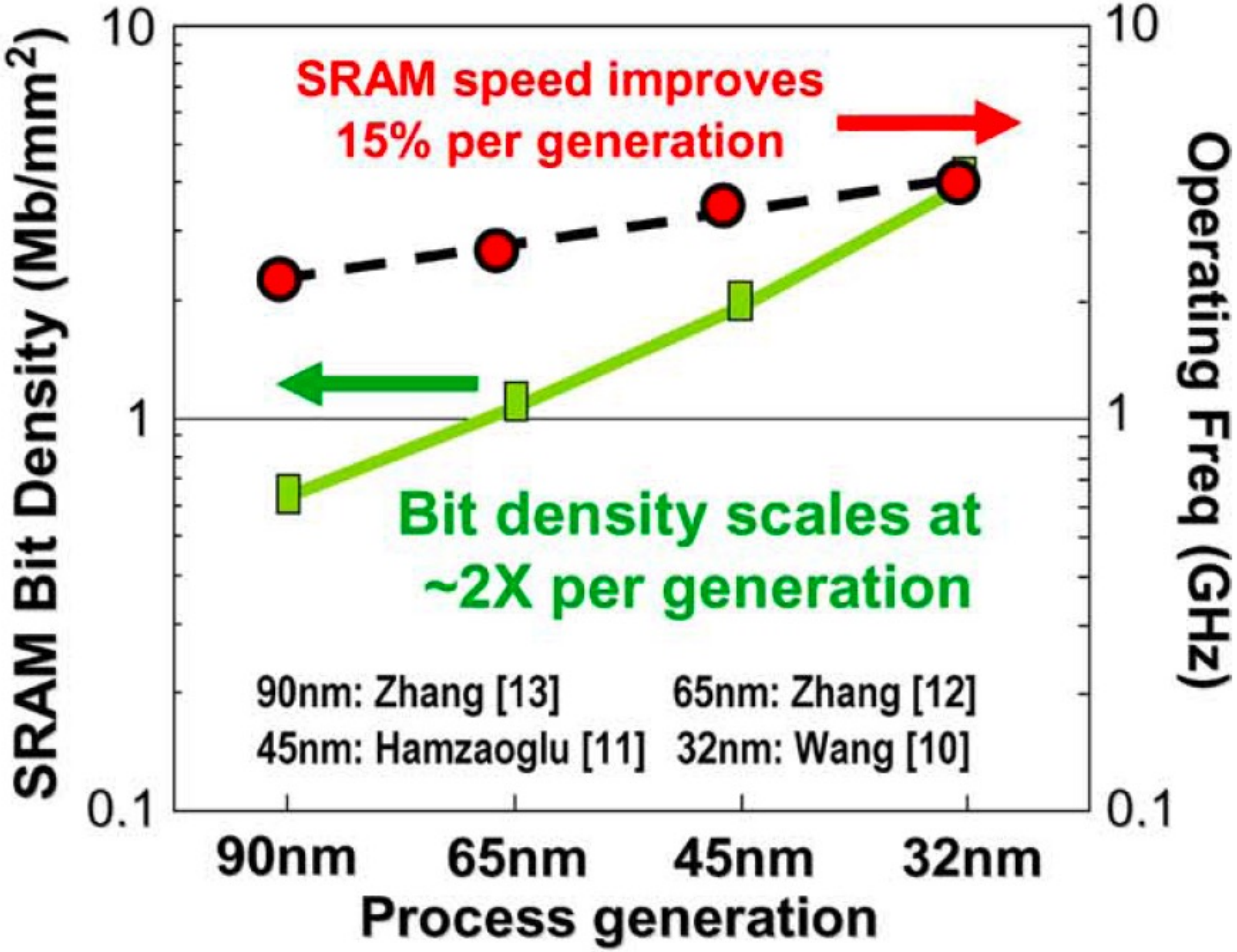
CMOS digital circuits are getting smaller ….



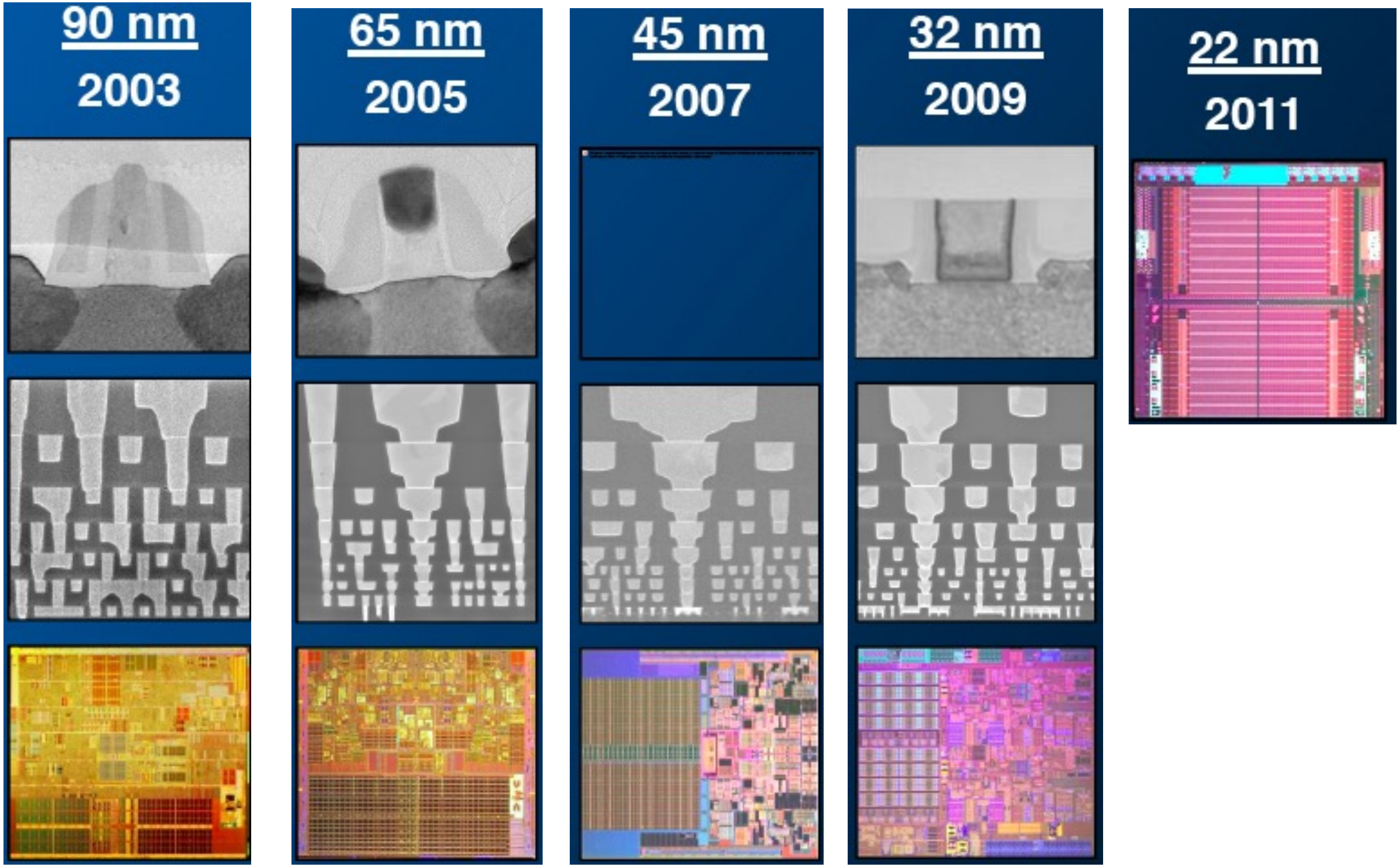
Source: Intel presentation by Singh, ISPD, 2011

Digital CMOS is also getting faster and denser



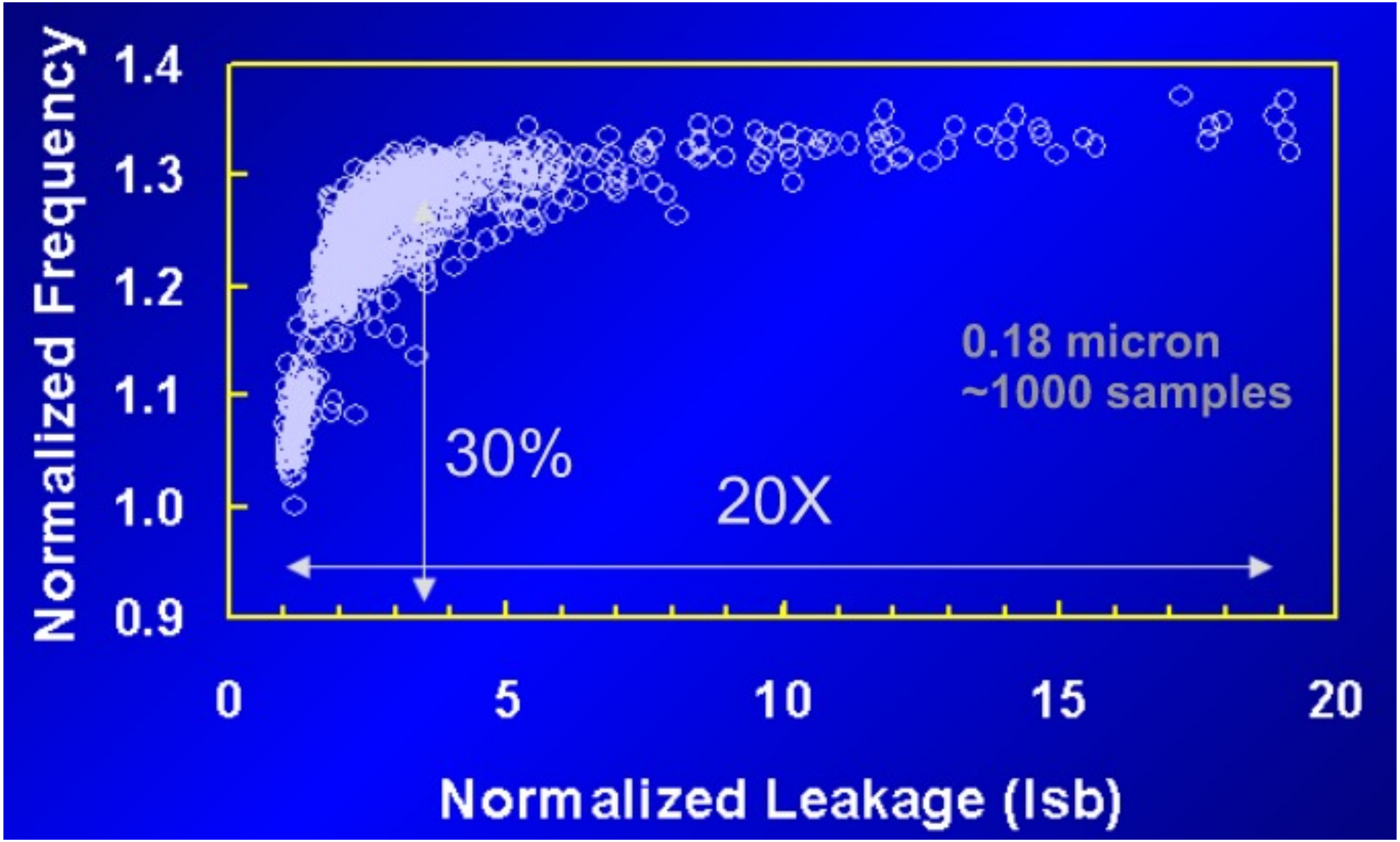
Source: Wang et al., "A 4.0GHz 291Mb Voltage-Saclable SRAM Design….", IEEE JSSC (45)1 2010.

… and it is happening every TWO years!



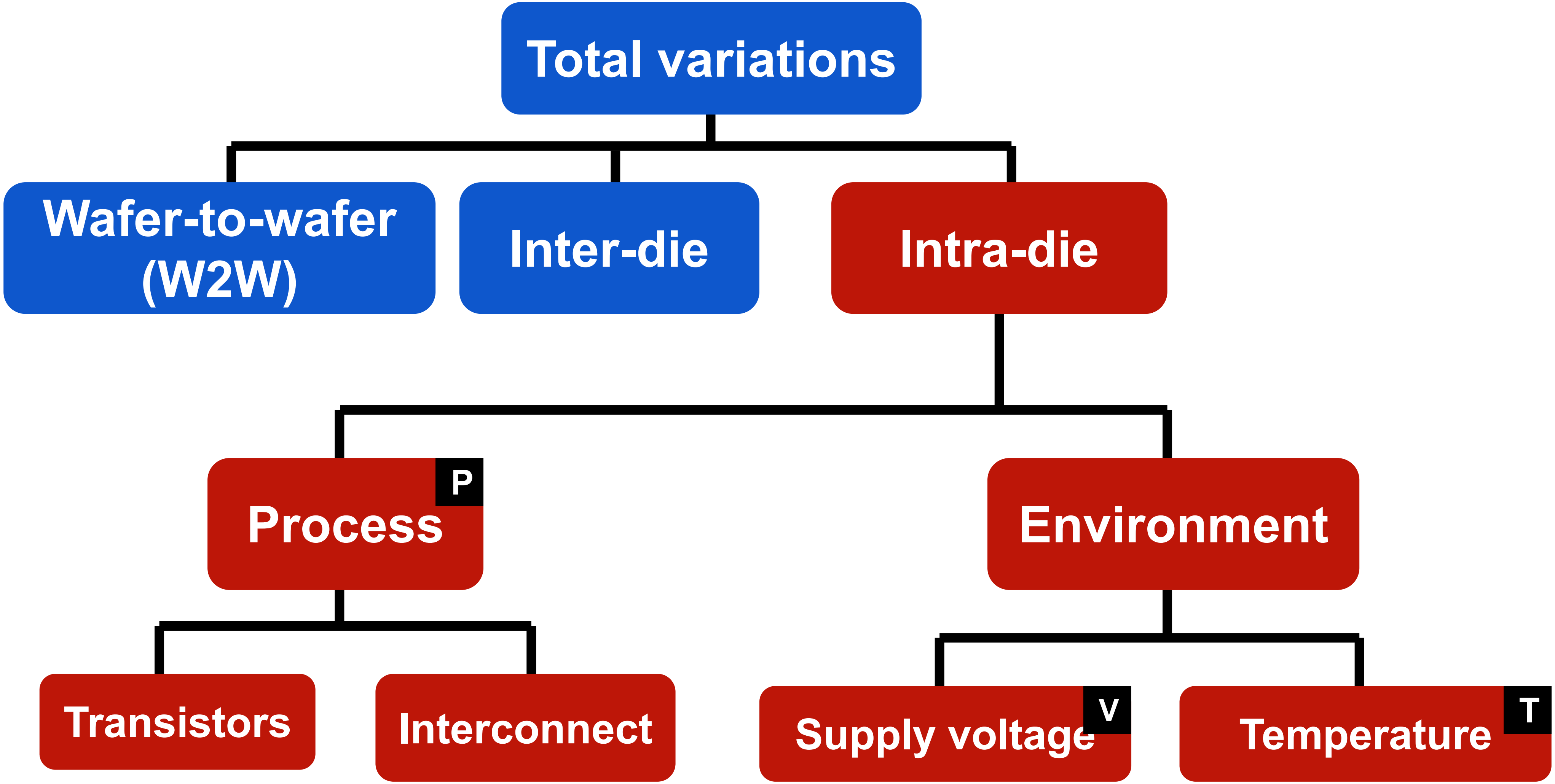
Source: Intel presentation by Singh, ISPD, 2011

But process scaling causes large variations



Source: Borkar et al. “Parameter Variations and Impact on Circuits and Microarchitecture”, DAC 2003

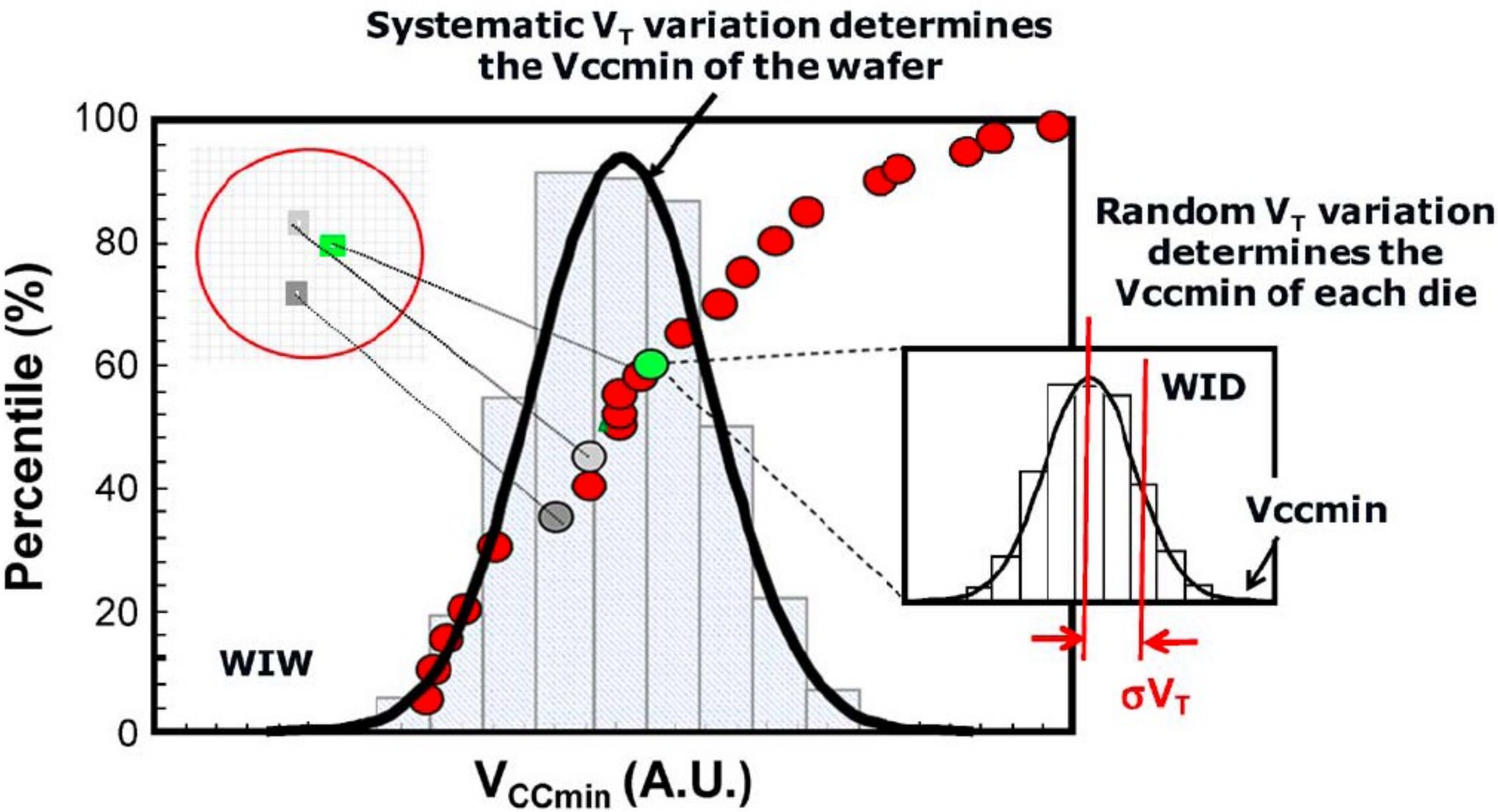
The top-level view on variations



Variations – P, V, T

- **Process** variations – due to the manufacturing process, particularly with technology scaling
- **Voltage** variations – due to changes in the supply voltage, the variations in IR-drops on board and on-chip, current fluctuations with changing of loads etc.
- **Temperature** variations – due to changes in environment temperature and/or on-chip temperature, e.g. local hotspots

Inter-die vs Intra-die variations

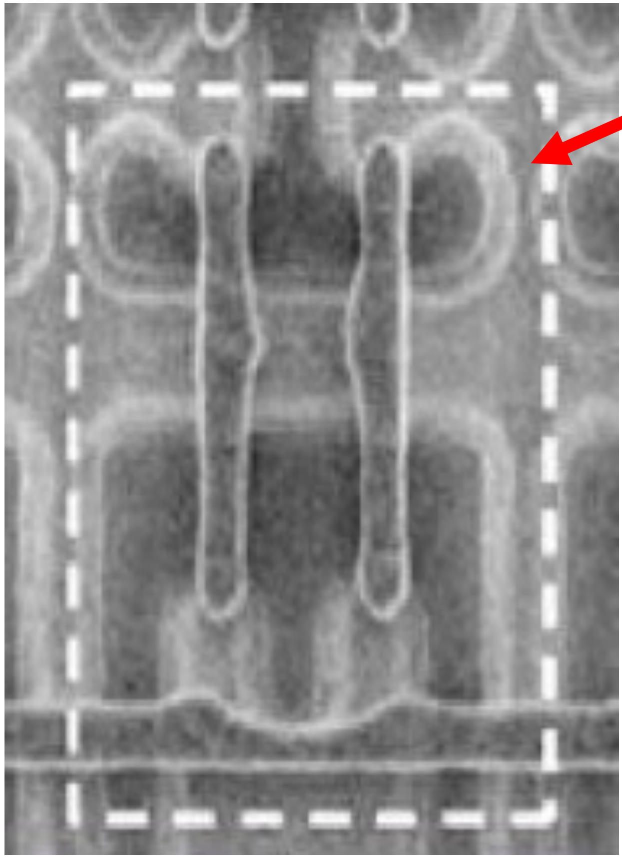


- Source: Wang et al., “A 4.0GHz 291Mb Voltage-Saclable SRAM Design….”, IEEE JSSC (45)1 2010.

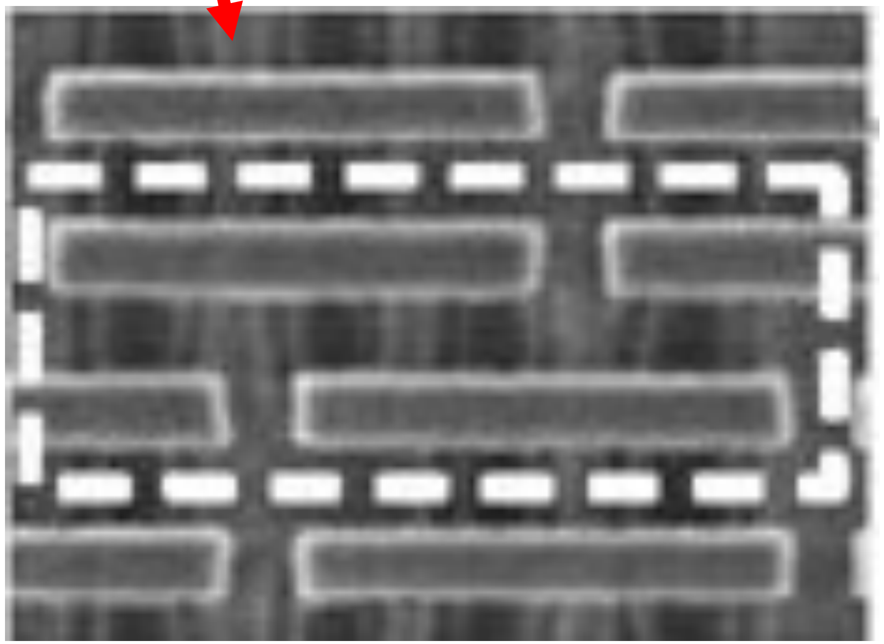
Sources of Process Variations – Old and New

- Historical Sources:
  1. Lithographical variations
  2. Line roughness
  3. Gate dielectric variations (e.g. oxide thickness, defects and traps)
- Emerging Sources (those becoming more important due to scaling):
  4. Random dopant fluctuation (RDF)
  5. Implants and annealing variations
  6. Strain variations (due to new process steps such as high-stress capping layers)
  7. Gate material granularity

Variations Sources – Line and feature roughness

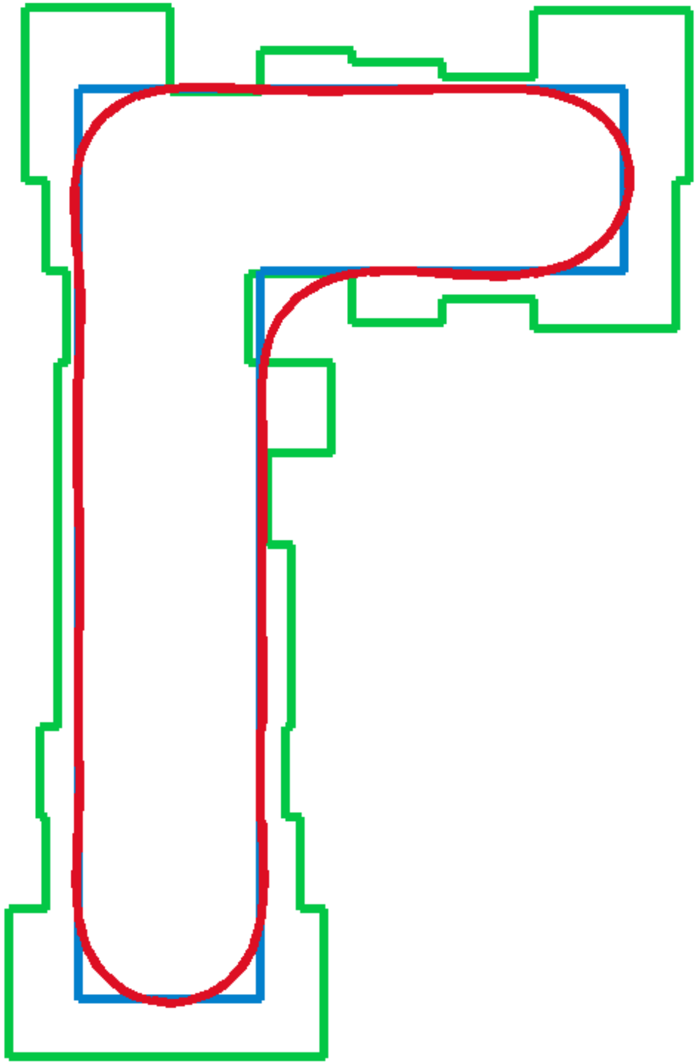


- 90 nm SRAM cell (left) with significant roughness and unevenness in lines features, corners and shapes
- Contrast this with a 32nm SRAM cell showing clear improvements in roughness, thus reducing variations



Source: Kuhn et al. "Process Technology Variations", T-ED (58), 8, 2011

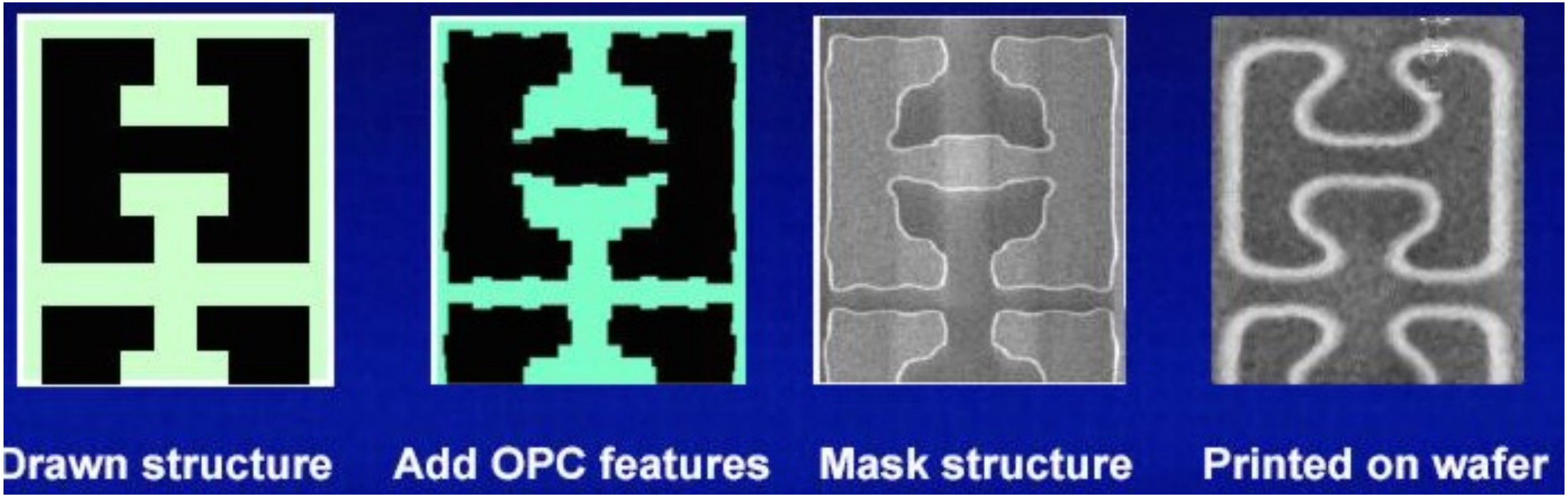
OPC reduces variations (1)



Source: Diagram from LithoGuy, wikipedia

- **Optical Proximity Correction (OPC)** is a technique to deliberately alter the mask with "roughness" in order to compensate for the non-ideal effect during the fabrication process.
- The **blue** shape is the desired feature, the **green** shape is the feature AFTER OPC is applied, the **red** shape is the fabricated feature.
- Corners are deliberated "bulged" in order to reduce the rounding after fabrication.
- OPC requires high cost in making masks and the electron beam needs to have small spot size and resolution. However, it provides significant improvement in reducing variations due to roughness.

OPC reduces variations (2)

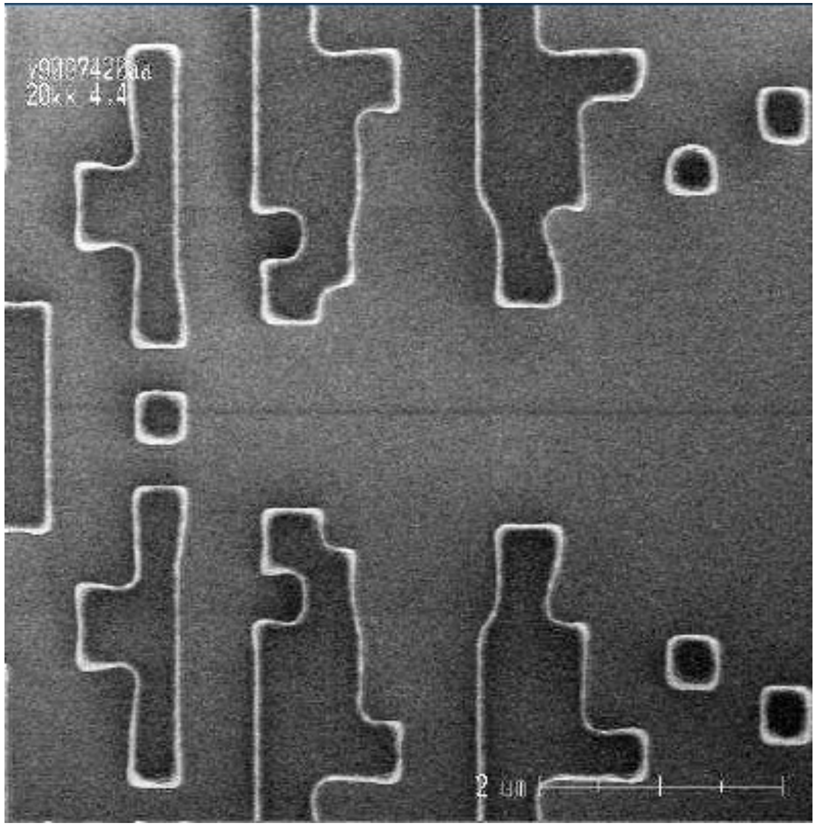


- OPC has been crucial from 130nm down to 32nm fabrication
- Moving to 22nm and below requires other techniques

Source: Intel presentation by Singh, ISPD, 2011

Other lithographical tricks – serif placement

Mask with serifs

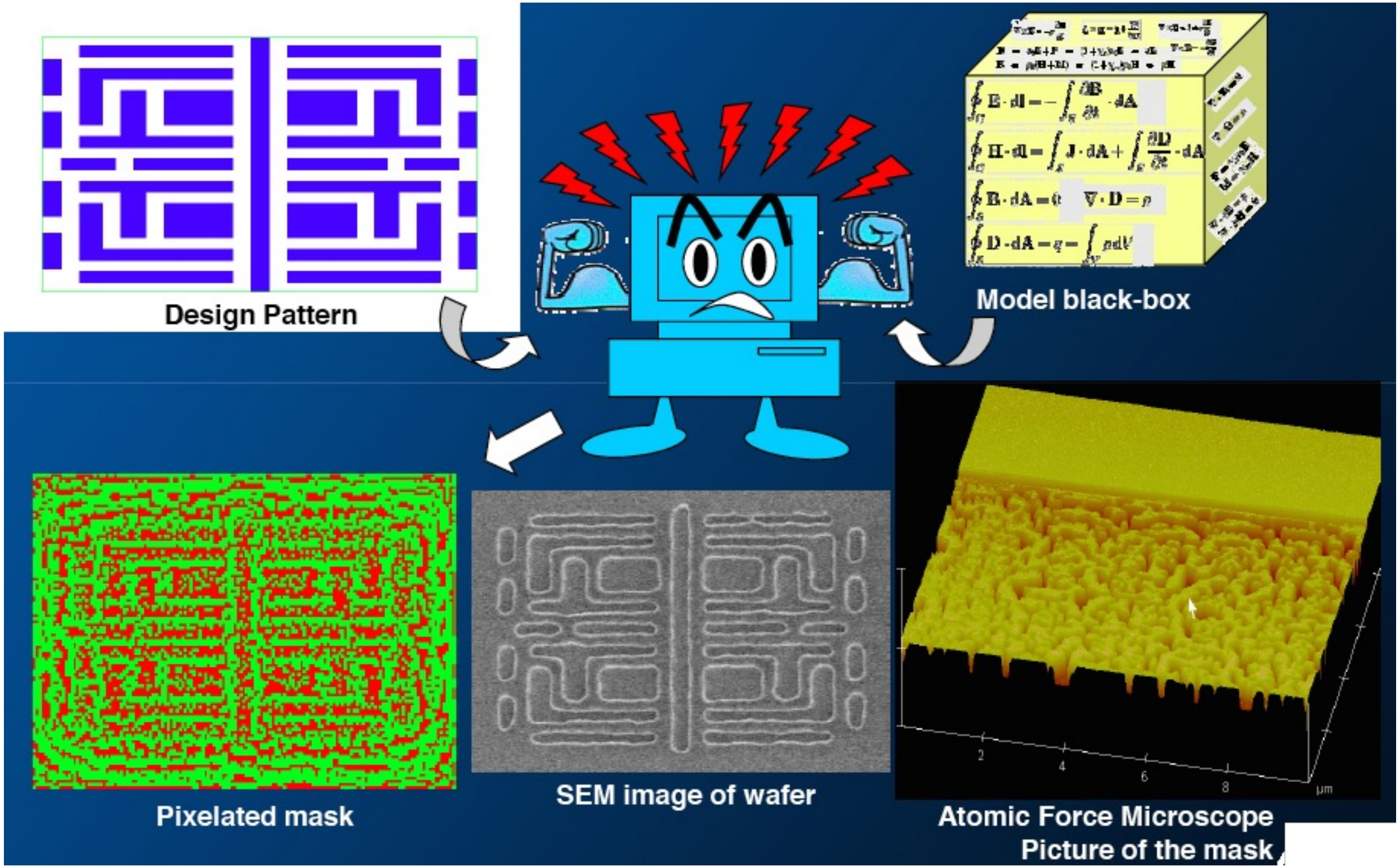


Fabricated

- Invented by Chen '96 Patent US570765
- Add little "tails" at corners

Source:Intel presentation by Singh, ISPD, 2011

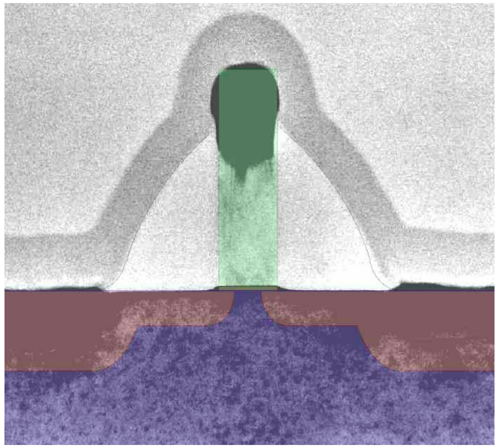
Other lithographical tricks – Pixelated phase mask



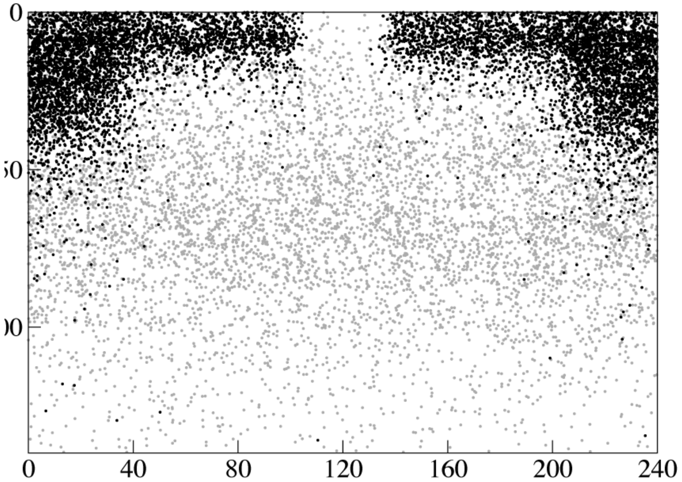
- Computational lithography – new techniques for further scaling

Source:Borodovsky et al., "Pixelated Phase Mask as Novel Lithography RET", Prc SPIE (6924) 2008 and Intel presentation by Singh, ISPD, 2011

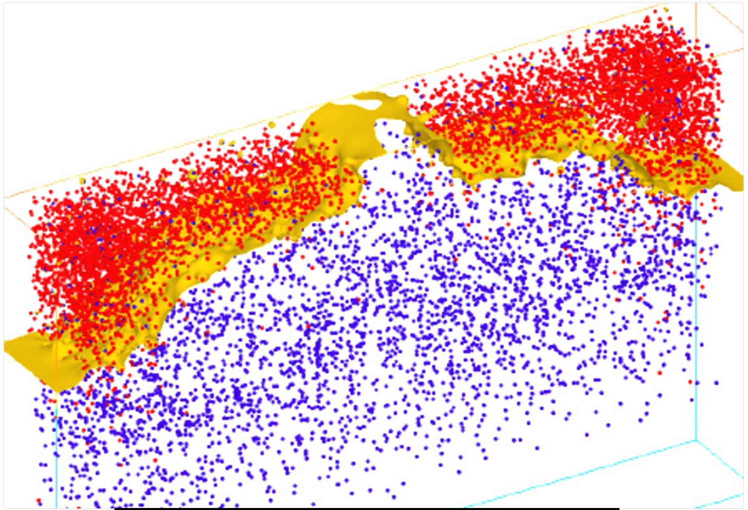
Process variations – random effects in doping



35nm Toshiba trans.



Dopant distribution

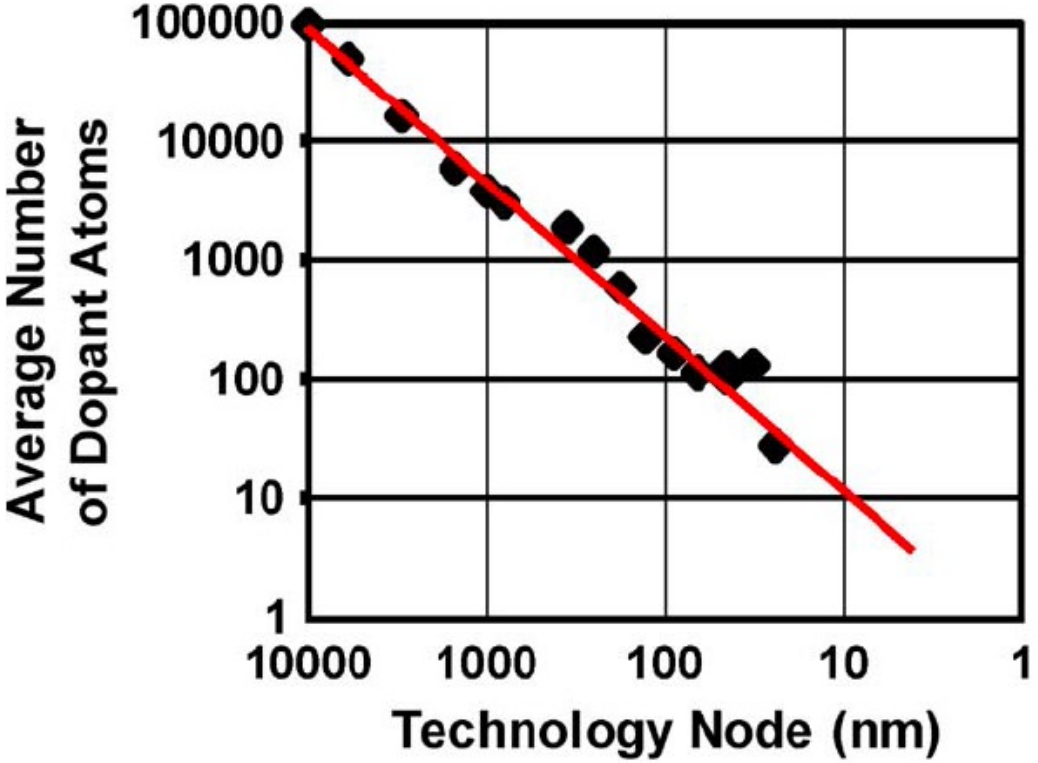


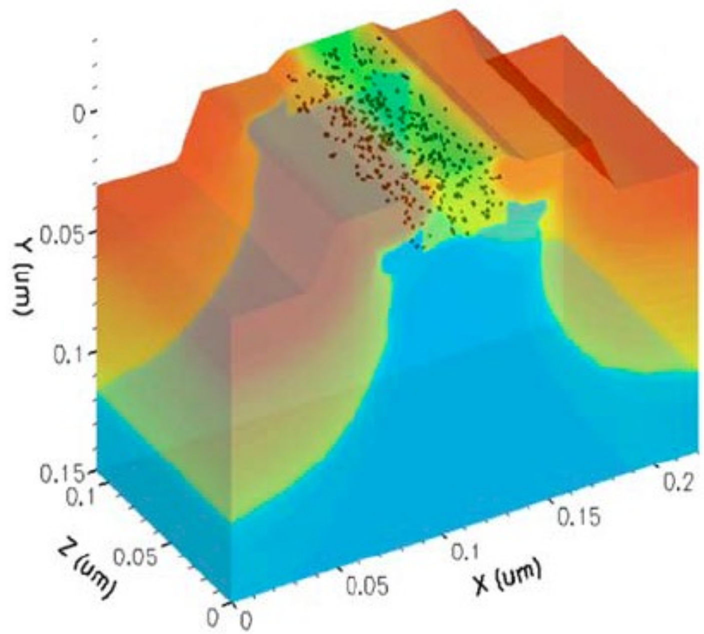
Dopant distribution

- Transistors are so small – each dopant atom counts
- Dopant location and density is random
- This causes random variations in device characteristic

Source: Asenov et al. "Integrated Atomistic Process and Device Simulatrion of Decananometre MOSFETS", SISPAD'02

Process variations – random effects in doping

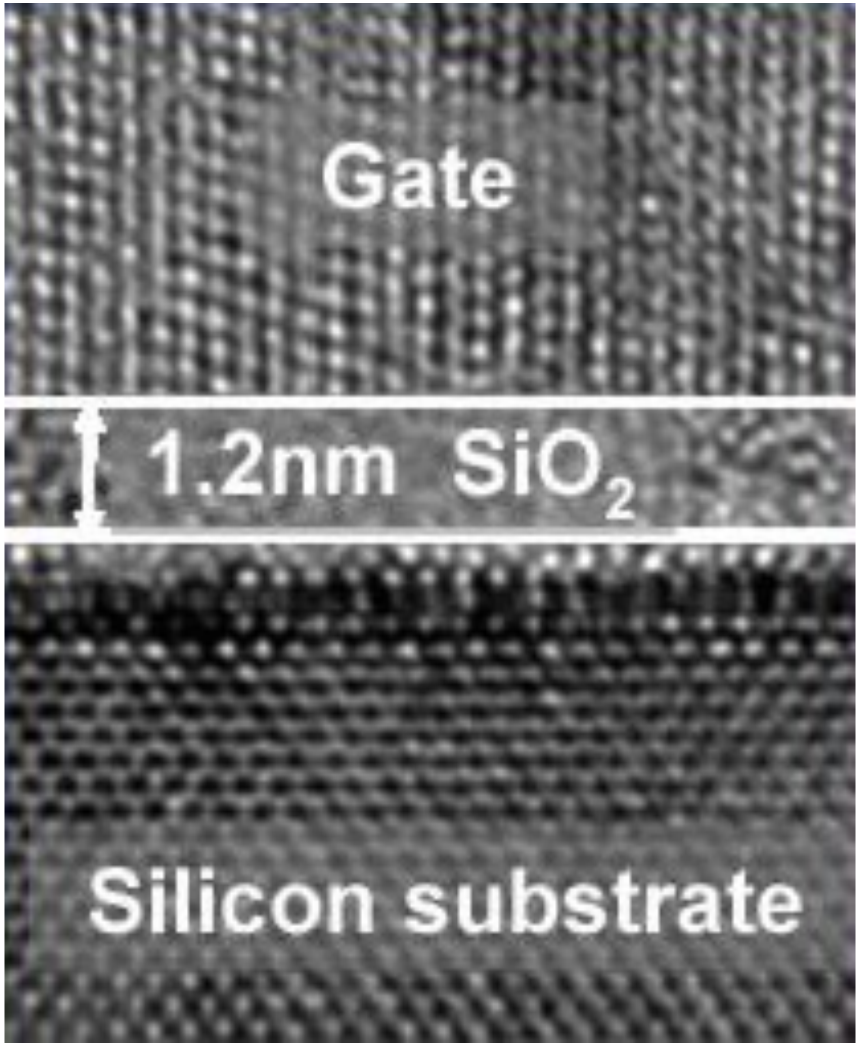




- Impact of random dopant fluctuation is increasing from generation to generation

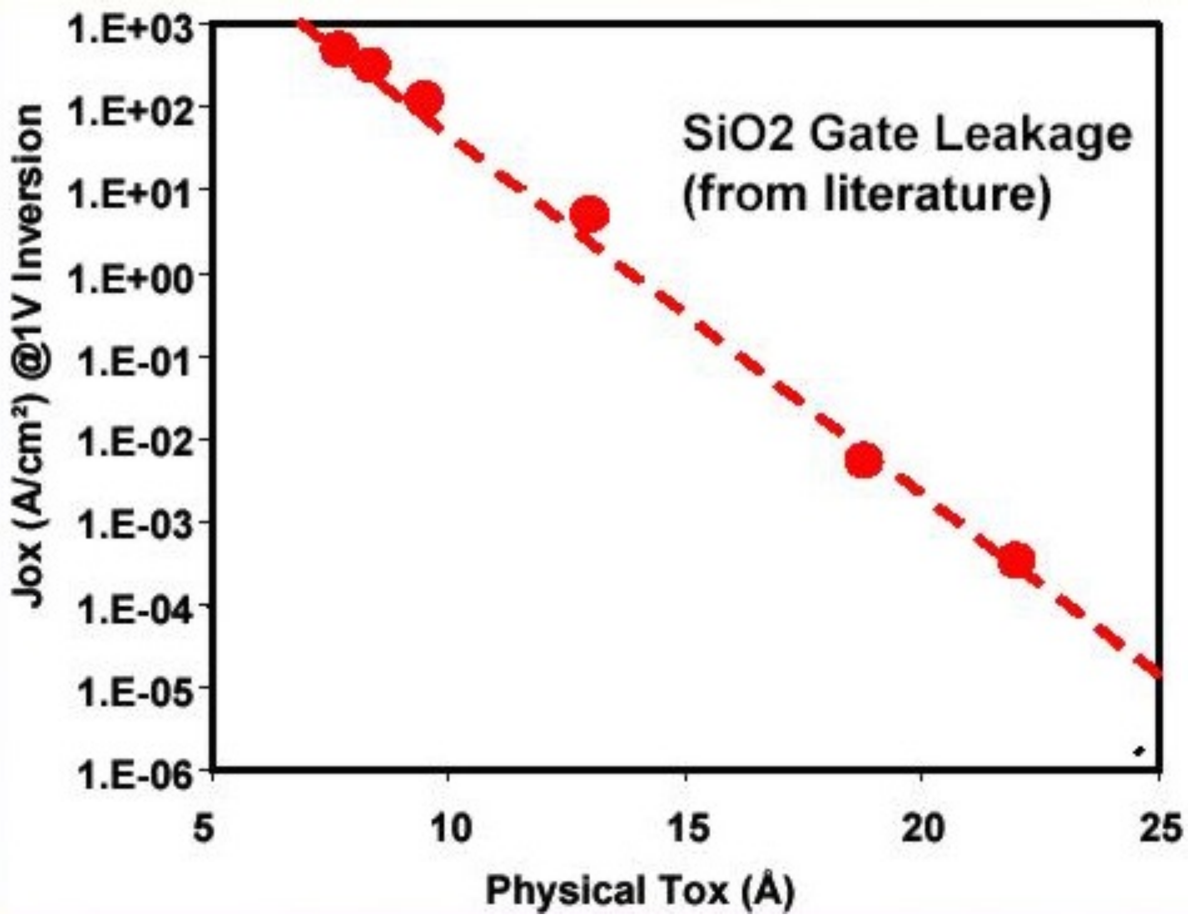
Source: Kuhn et al., “Managing Process Variation in Intel’s 45nm CMOS Technology” Intel Tech J. (12)2, 2008.

Process variations – very thin gate oxide layer



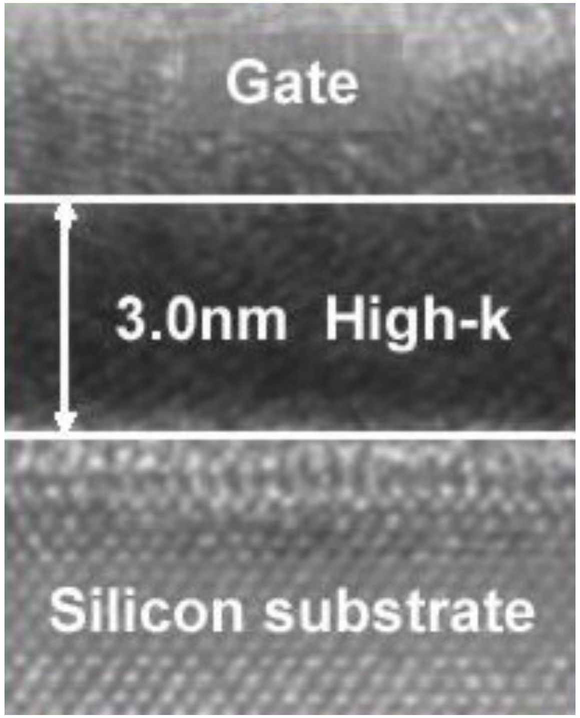
- Oxide thickness is now approaching the limit .
- It is now only around 5 atom thick!
- This thickness variation is another source of random variations.

Process variations – thinner oxide leads to larger leakage variations



- Thinner oxide gives faster transistors
- Unfortunately also result in higher leakage current
- Limit of SiO_2 thickness scaling is around 0.8nm

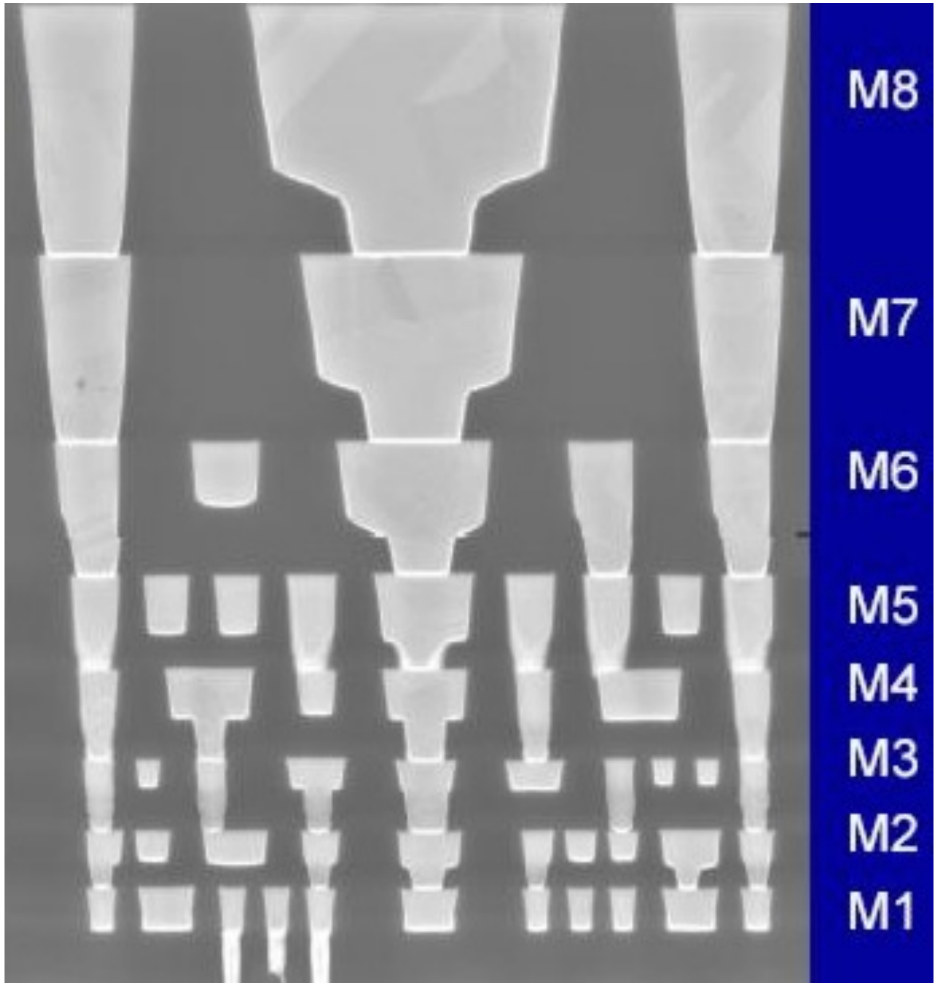
Process variations – High-k dielectric



- Where low leakage is required, use high-k dielectric materials
- Increase capacitance (hence field strength) by 60%
- This results in faster transistors
- Reduces leakage current (thicker), hence >x100 reduction in leakage
- Yet another new processing step

Source: Intel

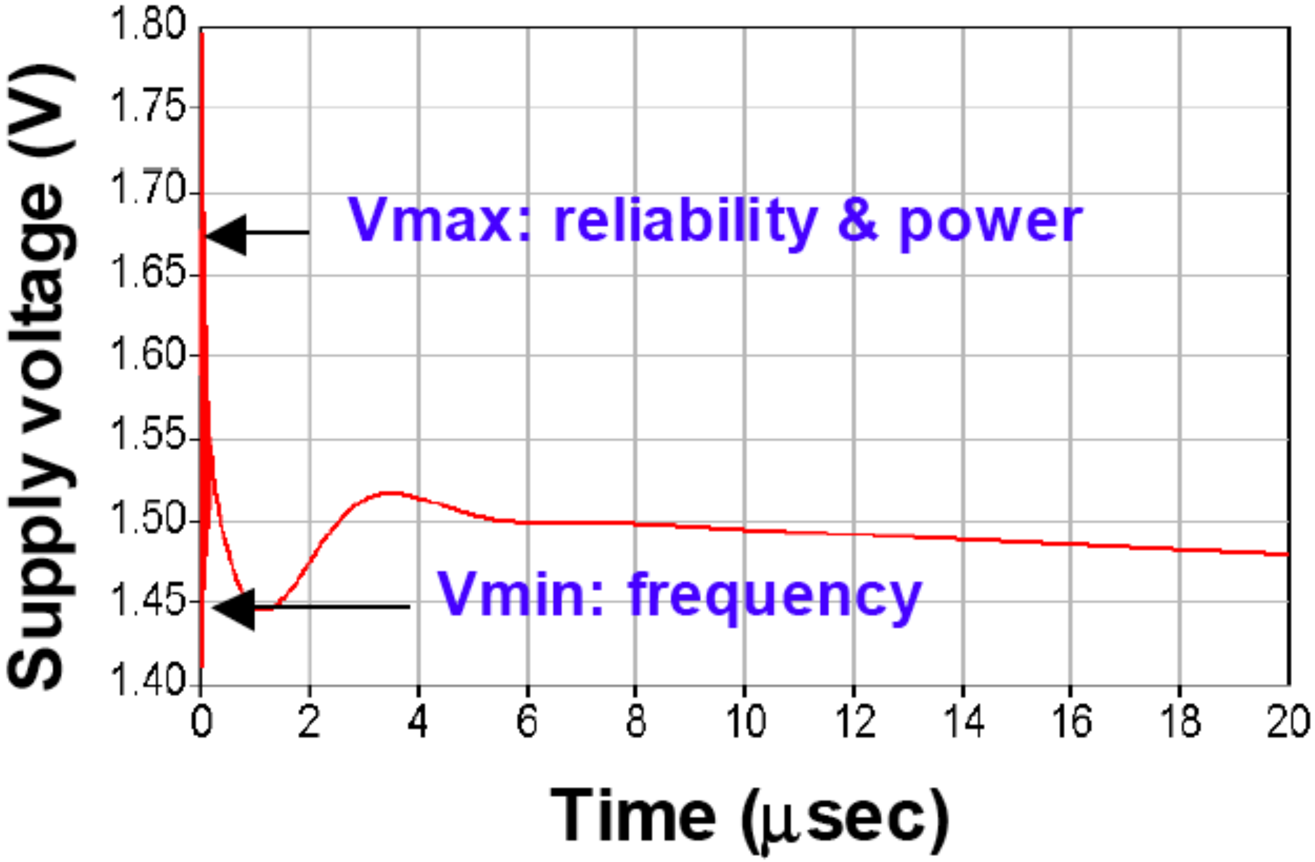
Process variations – Interconnect variations



- 65nm Intel process – 8 layer metal
- Low-k carbon doped oxide dielectric - reduced capacitance (~x0.7 of 90nm process)
- Line and via irregularities also increases random variations

Source: Intel

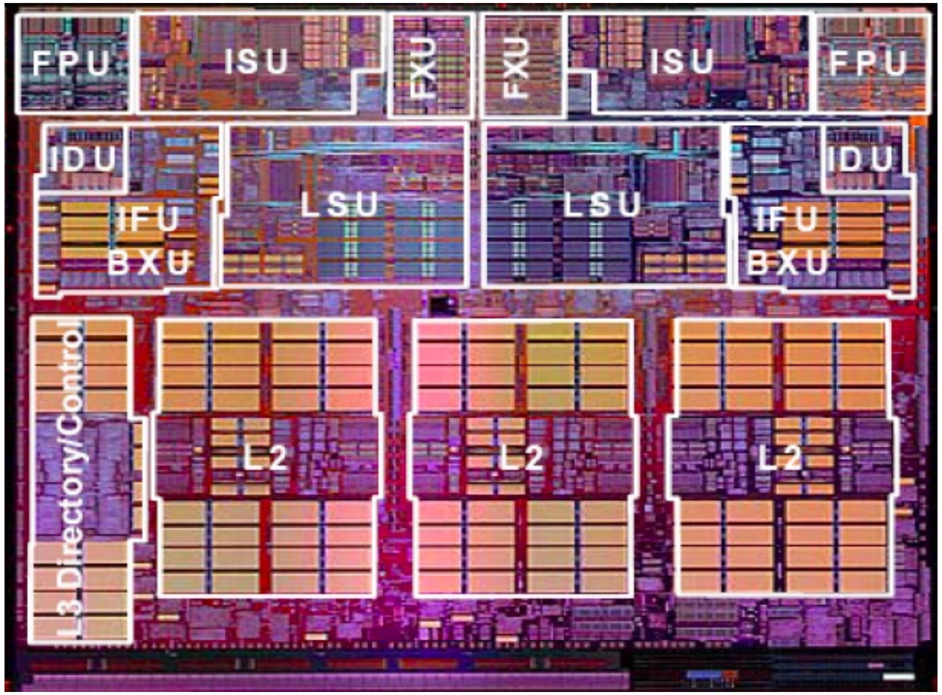
Supply voltage variations



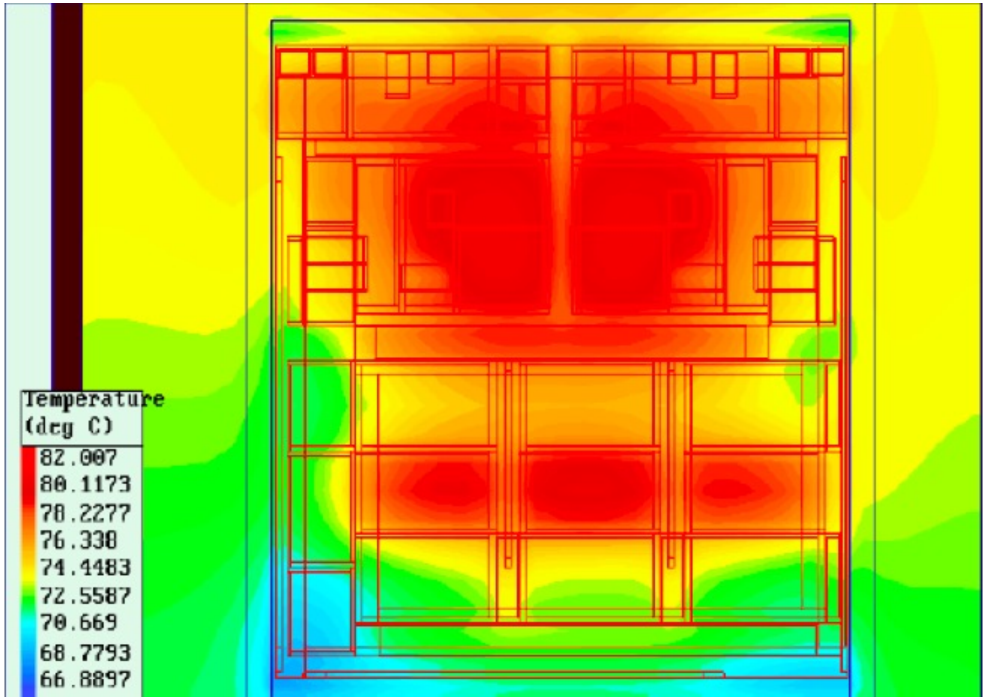
- Switching activities across the die result in uneven power and IR-drop
- Also result in local hot spots affecting Vth and leakage
- Vcc will continue to scale at around 15% with technology – lower than historical 30%
- This figure shows how supply voltage drops after chips started warming up

Source: Borkar et al. “Parameter Variations and Impact on Circuits and Microarchitecture”, DAC 2003

Temperature variations



Chip floorplan

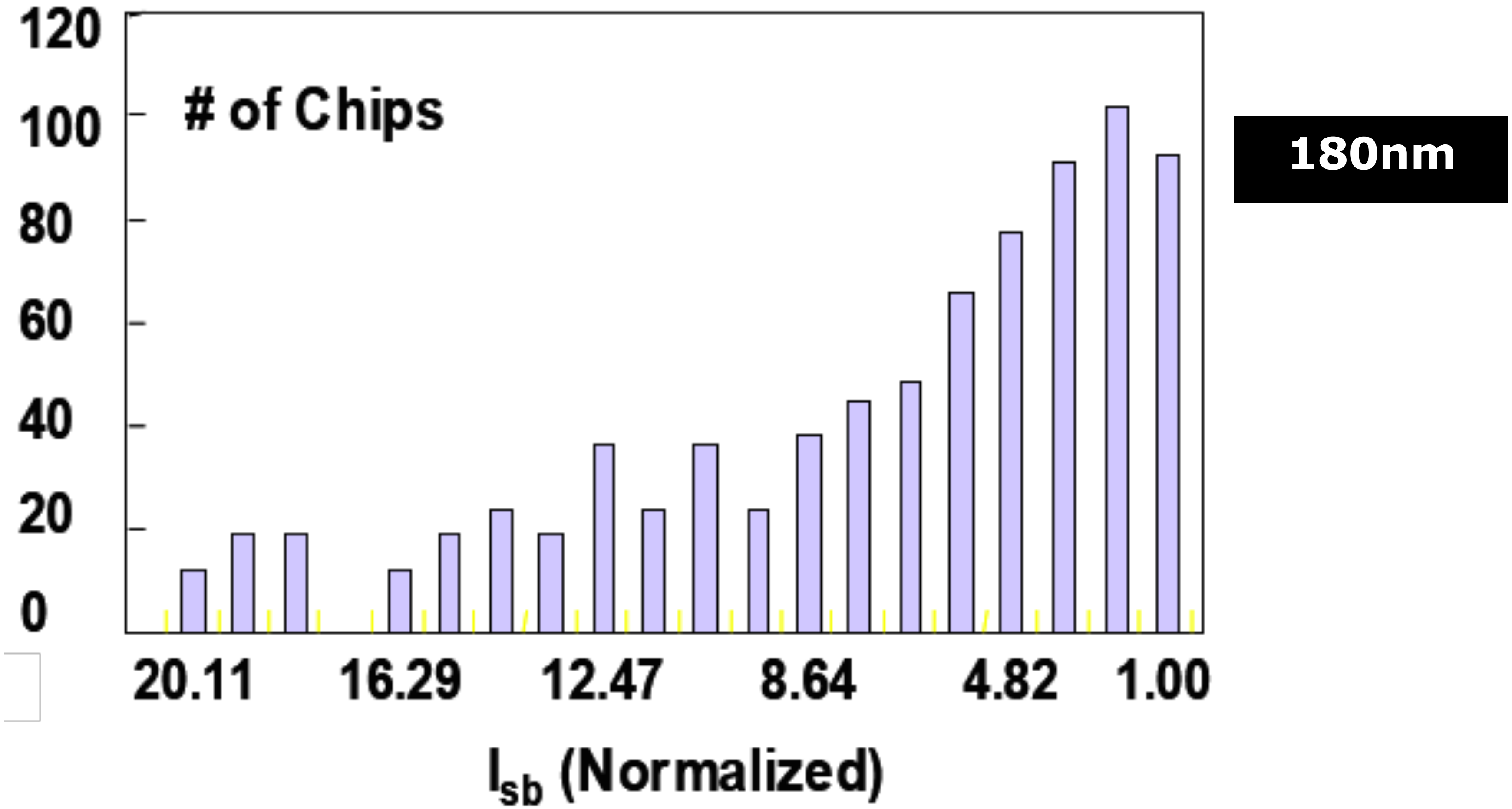


Thermal map

- Within-die temperature variations can be large – due to hot spots
- Here is a PowerPC chip with its temperature map during operation
- Increased temperature also degrades the hot parts of the chip faster – causing increase temporal variations

Source: IBM

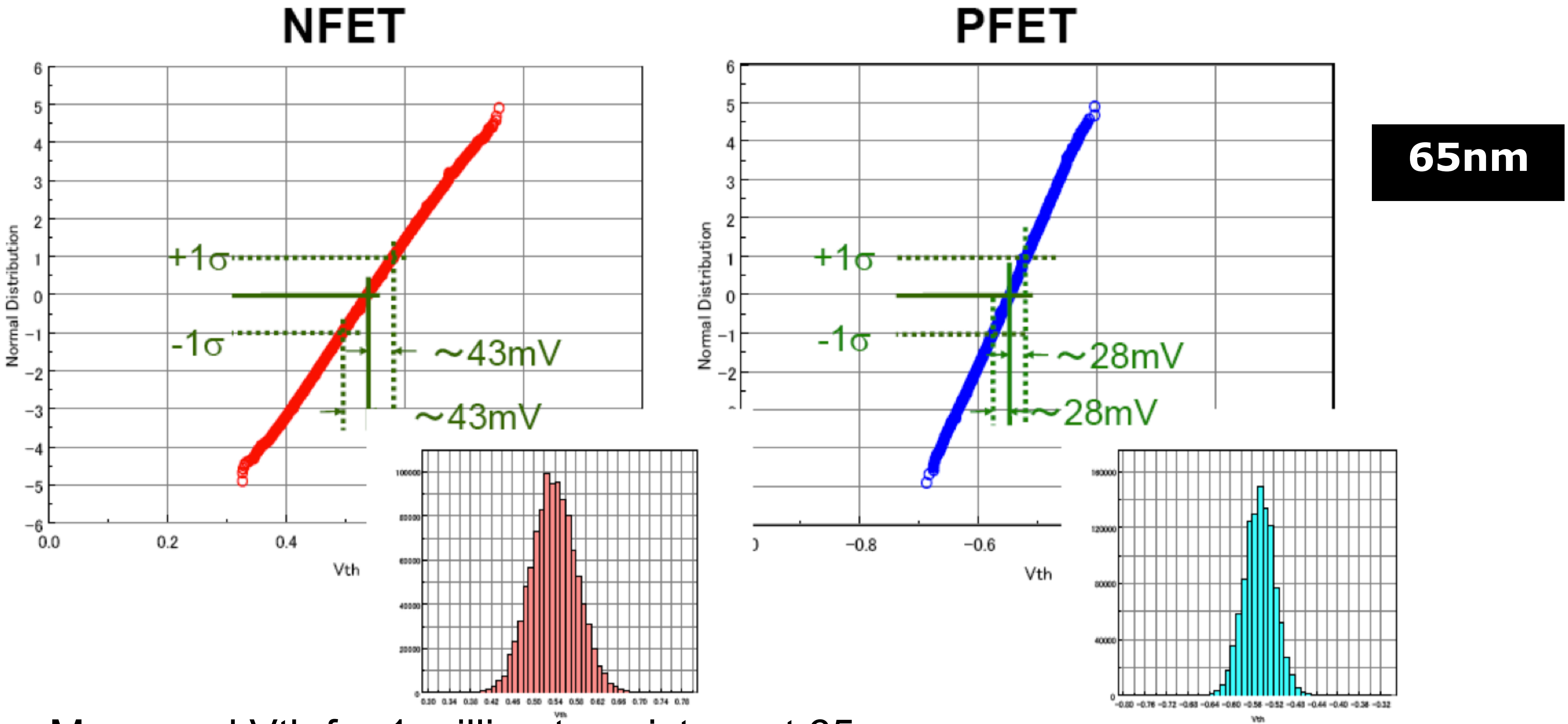
Impact of process variations on I_{sb}



- I_{sb} variations (substrate leakage current) can be as much as 20x!

Source: Borkar et al. "Parameter Variations and Impact on Circuits and Microarchitecture", DAC 2003

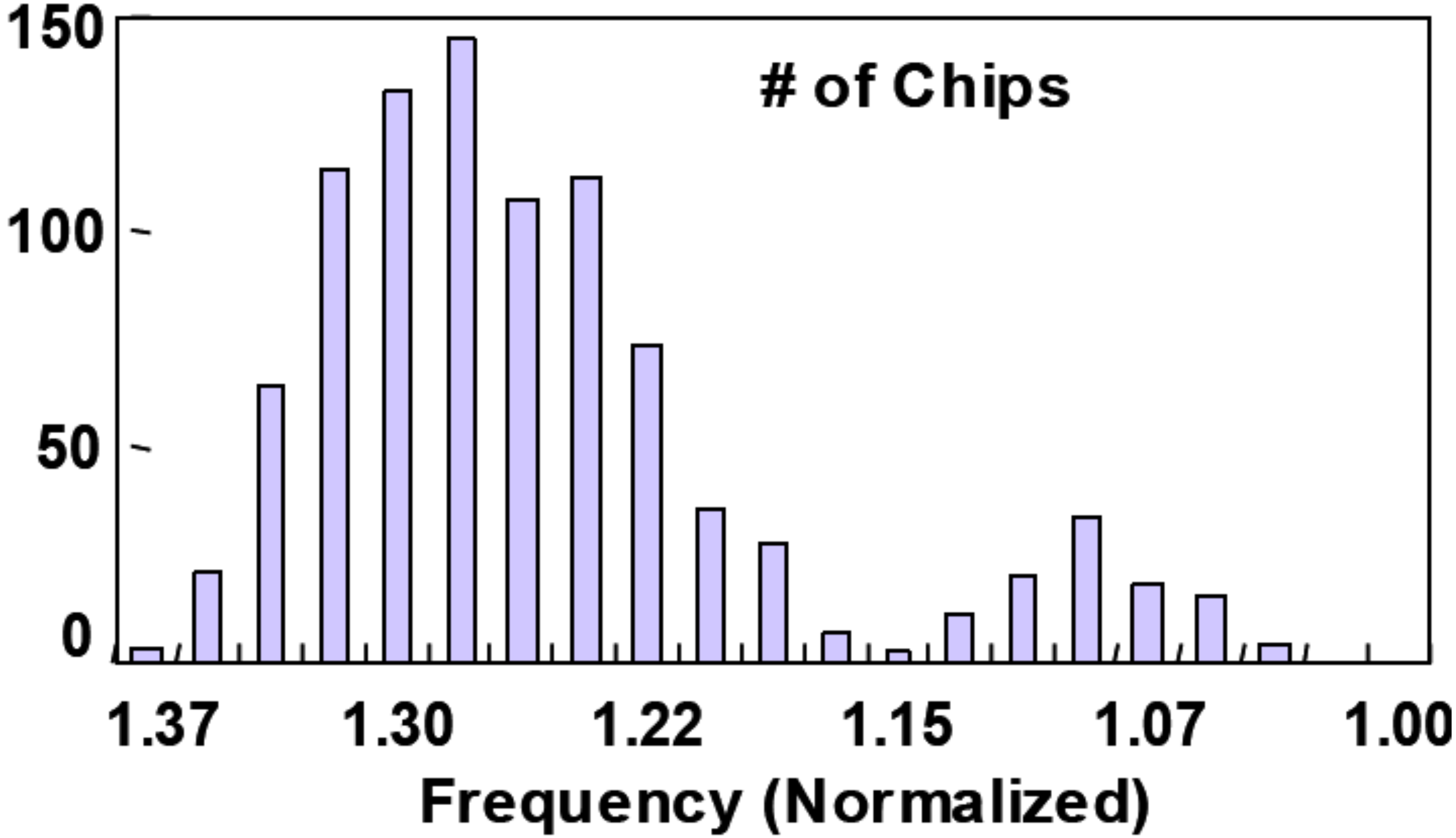
Impact of variations on V_{th}



- Measured Vth for 1 million transistors at 65nm
- Normal distribution up to ± 5σ
- NFET has larger variations than PFET

Source: Tsunomura et al. "Analyses of 5σ Vth fluctuation in 65nm MOSFETs", VLSI Symposium 2008

Impact of variations on operating frequency

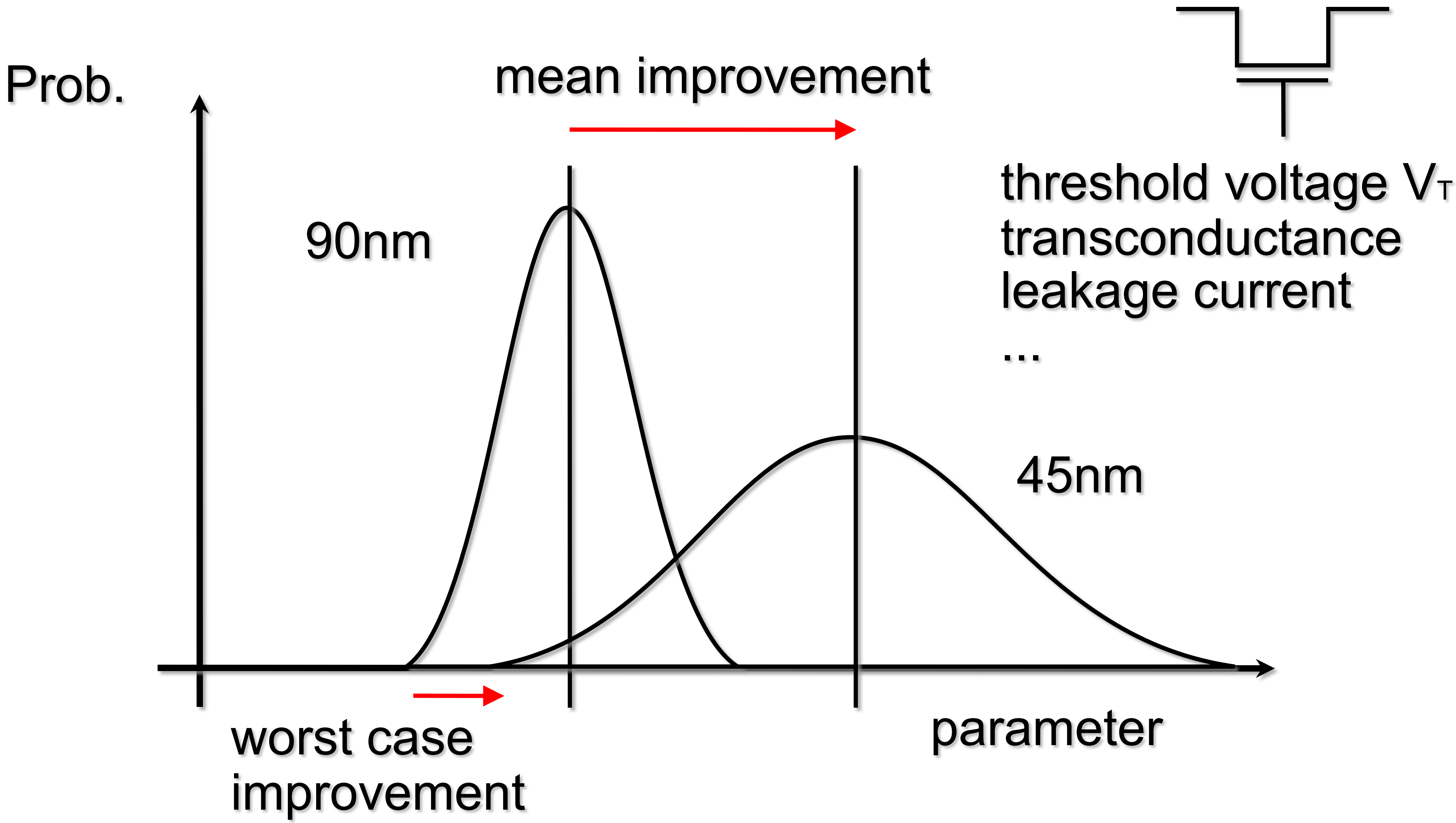


90nm

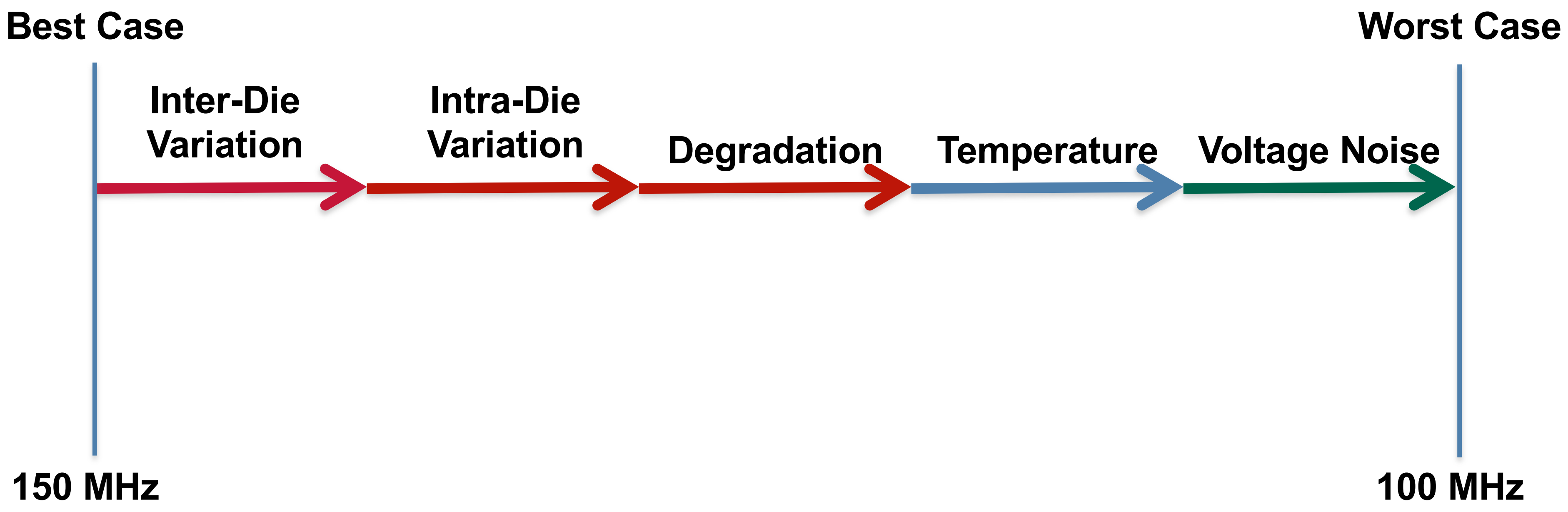
- As variation increases, the gap between the lowest and highest frequency could be as much as 40% for 90nm.
- The net effect is reduced the benefit of technology scaling on performance

Source: Borkar et al. "Parameter Variations and Impact on Circuits and Microarchitecture", DAC 2003

Variations reduces the benefit of technology scaling

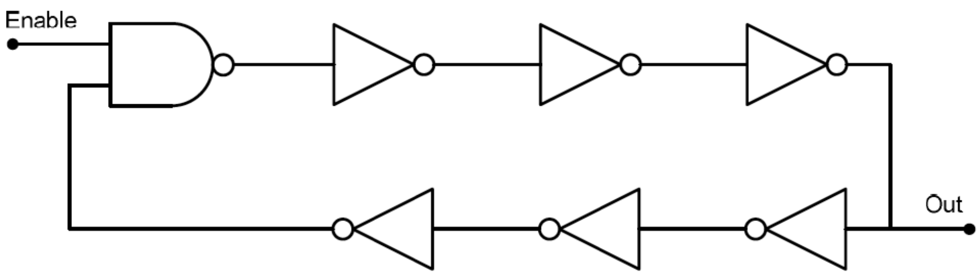


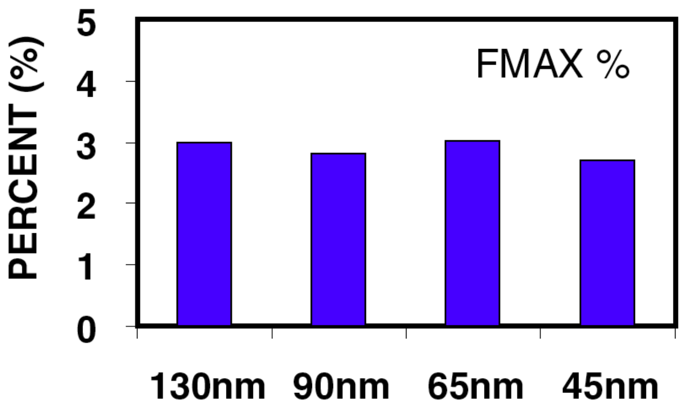
Timing Safety Margin Erosion

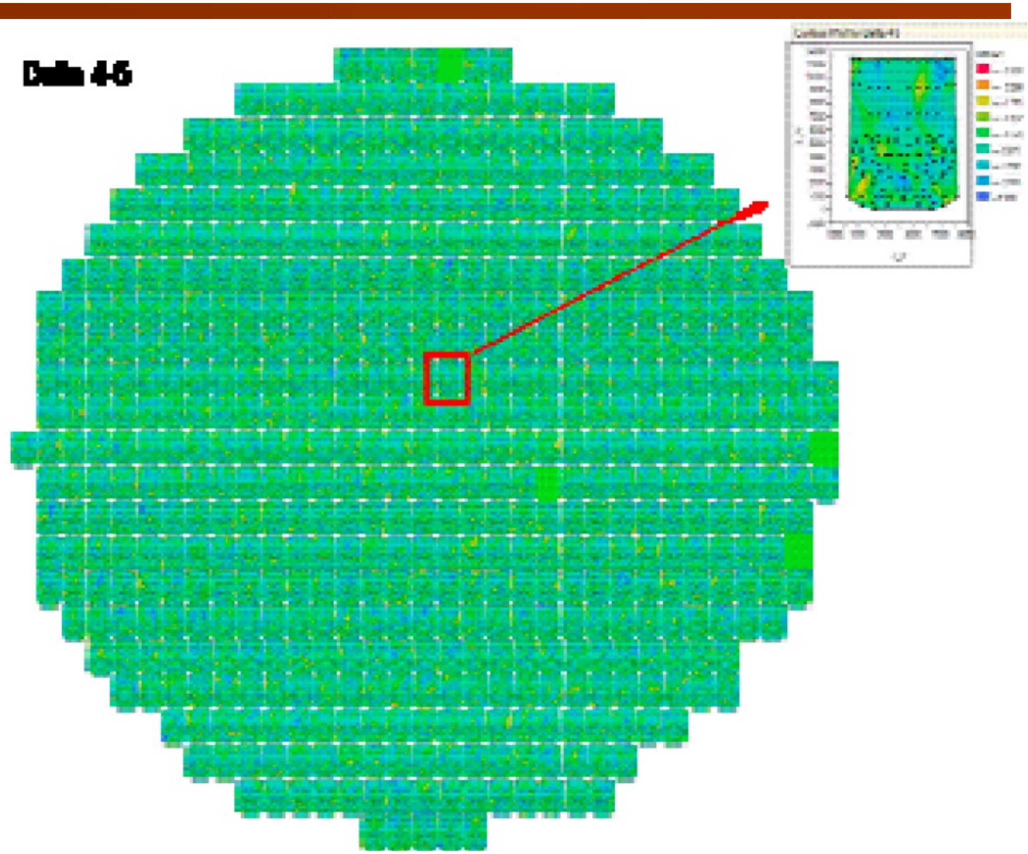


- The consequence of variations is to erode the possible maximum clock frequency from the best case scenario to cope with the worst case
- The first three factors are process related variations (P)
- The next two are temperature and voltage variations (T & V)
- Degradation is also a process variation where the transistors age over time, and gets slower

Ring Oscillator – traditional way of measuring variations







- Using ring oscillators populated with-in die and around the wafer, it is possible to measure variation indirectly through the maximum oscillator frequencies.
- Impact of variations remain around 3% as reported by Intel on their 130nm to 45nm process nodes

Source: Kuhn et al., "Managing Process Variation in Intel's 45nm CMOS Technology" Intel Tech J. (12)2, 2008.

Cyclone III (65nm): Carry chain test movie